

CLAVE(S): TP P55

**BOMBAS PARA MANEJO DE
CONDENSADO**

CLIENTE: *

TITULO DEL PROYECTO: *

NÚMERO DE TRABAJO: *

NÚMERO DEL EQUIPO: TP P55

SERVICIO DEL EQUIPO: BOMBAS PARA MANEJO DE CONDENSADO

NÚMERO DE SERIE: AL FABRICAR

NO. REQ / ESPEC: *

NO. DE ORDEN DEL COMPRADOR: *

COMENTARIOS:

HOJAS DE DATOS					
NO. MAQUINARIA	ATT	ITEM NO.	ATT	ITEM NO.	ATT
BOMBA					
MOTOR					
ENGRANE					
TURBINA					

0	18/06/2026	PARA APROBACION			AMB	RGG	MMC		
Rev.	FECHA	DESCRIPCIÓN			Por	Revisado	Aprobado	Por	
		HOJAS DE DATOS DE BOMBA CENTRIFUGA			NO. HOJA DE DATOS				
					BIM-HDT-TP P55				
					HOJA:	1 de 10			
					FOGV.01.03				

HOJAS DE DATOS DE BOMBA CENTRIFUGA

Cliente: *
 No. de H.D.: BIM-HDT-TP P55
 No. de serie: AL FABRICAR
 No. REQ / ESPC: *
 No. O.C.: *
 Rev: 0 Hoja: 2 de 10

APLICABLE A : COTIZACIÓN
 PARA : *
 SITIO : *
 NO. DE REQ: *
 FABRICANTE: BIMSA TAMAÑO DE BOMBA: 8 FMV-501 MODELO : 1305L No. SERIAL: AL FABRICAR
 No. DE EQUIPO: TP P55 CANTIDAD: 2 (DOS) OPERAN EN:
 TIPO DE ACCIONAMIENTO: ☒ MOTOR ☐ TURBINA
☒ Eléctrico ☐ Combustión interna

CARACTERISTICAS DEL LIQUIDO

UNIDADES	MAXIMO*	MINIMO*	NOTA	SERVICIO:	N/A
TIPO DE LIQ. O NOMBRE:	CONDENSADO			*SI ES INTERMITENTE, NO. DE ARRANQUES / DIA	
PRESION DE VAPOR: kg/cm ² -a				CORROSION DEBIDO A (6.12.1.9) :	N/A
GRAVEDAD ESPECÍFICA:	1.000		*Máx. &Min. Los valores se refieren únicamente a las propiedades listadas	EROSION DEBIDO A (6.12.1.9) :	N/A
CALOR ESPECIFICO: kcal/kg °C				CONCENTRACION H2S (ppm) (6.12.1.12) :	N/A
VISCOSIDAD: cP	1.000				

CONDICIONES DE OPERACIÓN (6.1.2)

UNIDADES	MAXIMO	NOMINAL	NORMAL	MINIMO	
DATO DE NPSHa:	L.C IMPULSOR				
TEMPERATURA DE BOMBEO: °C		AMBIENTE			
FLUJO: m ³ /h		31.82			
PRES. DE DESCARGA: (6.3.2): kg/cm ² -g	-	2.92			
PRESION DE SUCCION: kg/cm ² -g	-	0.00			
PRESION DIFERENCIAL: kg/cm ²		2.92			
CARGA DIFERENCIAL: m		29.26			
NPSHA : m		2.90			
POTENCIA HIDRAULICA: kW					

NOTAS PRESION DE DECARGA (Pd)

SITIO Y DATOS DE SERVICIO (6.1.2)

LOCALIZACION:	AGUA DE ENFRIAMIENTO:
EXTERIOR	TEMP. DE SUMINIST.: °C MAX. TEMP. DE RET. PERMIS. °C
MONTADO EN: (X) TROPICALIZACION REQ'D	PRESION NORMAL: kg/cm ² -g PRES. DE DISEÑO: kg/cm ² -g
CLASIFICACION DE AREA ELECTRICA: NO PELIGROSO	CONCENTRACION DE CLORURO: ppm DESIGN T: °C
CLASE: I GRUPO B C & D DIVISION : 2	AIRE DE INSTRUMENTOS MAX.: kg/cm ² -g MIN.: kg/cm ² -g
DATOS DE SITIO: COD. DE TEMP.: T3	MECH DESIGN: MAX.: kg/cm ² -g MIN.: kg/cm ² -g
ELEVACION (M.S.N.M) m BAROMETRO: kg/cm ² -a	VAPOR:
RANGO DE TEMP. AMBIENTE MIN./MAX. / °C	ACCIONADOR CALENTADOR
HUMEDAD RELATIVA: MIN. / MAX. : / %	TEMP: °C MAX. : MIN. :
CONDICIONES INUSUALES:	PRESION: kg/cm ² -g MAX. : MIN. :
CONDICIONES A UTILIZAR:	
ELECTRICIDAD: ACCIONADOR CALENTAMIENTO CONTROL PARO	
VOLTAGE: 460 120 24 24	
FASE: 3 1	
HERTZ: 60 60	

COMPORTAMIENTO

ACCIONADOR (7.1.5)

NO. DE CURVA PROPUESTA: 8 x 6 - CB RPM 1760	TIPO DE ACCIONADOR: MOTOR DE INDUCCION
NO. DE CURVA DE PRUEBA: *	ENGRANE: NO
DIAM. IMPULSOR: NOMINAL: 147.64 MAX: 147.64 MIN: 130.18 mm	REQ. VELOCIDAD VARIABLE: NO
POTENCIA NOM.: 3.56 kW EFICIENCIA: 71 %	VELOCIDAD DE VELOCIDAD VARIABLE:
FLUJO NOMINAL EN CURVA BEP: (al diámetro nominal del impulsor) 33.86 m ³ /h	OTRO: 460 / AISLAMIENTO F / 1.15 F.S.
FLUJO MIN. TERMICO 11.93 m ³ /h ESTABLE 17.05 m ³ /h	FABRICANTE: US
REGION DE OPERACIÓN PREFERENTE: (6.1.12) 23.70 a 40.64 m ³ /h	POTENCIA EN PLACA: 7.456 kW
REGION DE OPERACION PERMISIBLE: 17.05 a 50.48 m ³ /h	NOMINAL RPM: 1800
CARGA MAX. @ IMPULSOR NOMINAL: 33.71 m	RPM A CARGA NOMINAL: 1800
POTENCIA MAX. @ IMPULSOR NOMINAL: (6.8.9) 4.15 kW	ARMAZON O MODELO: TEFC
NPSHR a FLUJO NOMINAL: 1.40 m	ORIENTACION: VERTICAL
LC DE BOMBA A BASE: 9 m	LUBRICANTE: GRASA
MARGEN DE NPSH A FLUJO NOMINAL: 1.50 m	BALEROS TIPO: BALEROS ANTIFRICCIÓN
VELOCIDAD ESPECIFICA: m ³ /s, RPM, m 30.62	RADIAL: BOLAS / 6309-J/C3
LIMITE DE VELOCIDAD ESPECIFICA DE SUCCION: 213.06	EMPUJE: BOLAS / 6208-J/C3
VELOCIDAD ESPECIFICA DE SUCCION: (6.1.9) m ³ /s, RPM, m 123.44	METODO DE ENCENDIDO: A TENSION PLENA
NIVEL EST. MAX. DE PRES. DE RUIDO PERMI.: (6.1.14) @1.5 m 65 dBA	VER HOJA DE DATOS DEL ACCIO.:
NIVEL EST. MAX. DE POTENCIA DE RUIDO PERMI.: (6.1.14) @1.5 m	

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		HOJAS DE DATOS DE BOMBA CENTRIFUGA		* No. de H.D.: BIM-HDT-TP P55 No. de serie: AL FABRICAR No. REQ / ESPEC: * No. O.C.: * Rev: 0 Hoja: 3 de 10				
1	Nota	CONSTRUCCION						REV
2		BOMBA TIPO API: VS6 [Basado en definiciones del API 610]				MONTAJE DE CARCASA: VERTICAL		
3						TIPO DE CARCASA: (6.3.10) DIFUSOR		
4		BOQUILLAS DE CONEXIÓN:				DISPOSITIVO OH3 REQ. PARA IZAJE - CARGA (9.1.2.6)		
5		TAMAÑO	CARA	RANGO	POSICION	CAPACIDAD DE LA CARCASA A PRESION:		
6		SUCCION 8"	R.F.	150 #	LATERAL	MAWP: (6.3.6) 19.33	kg/cm²-g @ °C	
7		DESCARGA 6"	R.F.	150 #	LATERAL	P. HIDROSTAT.: (8.3.2.6) 29.88	kg/cm²-g @ °C	
8		CONEXIONES AUX. CARCASA A PRESION: (6.4.1.2)(6.4.3.1)(6.4.3.2)(6.4.3.12)						
9		NO.	TAMAÑO	TIPO	CARA	RANGO	POSICION	
10		BALANCE/FUGA						
11		DREN 1	3/4	SW	R.F.	150 #	HORIZONTAL	
12		VENTEO 1	3/4	SW	R.F.	150 #	HORIZONTAL	
13		MEDIDOR DE PRES.						
14		MEDIDOR DE TEMP.						
15		PRECALENTAMIENTO:						
16		VALVULA DE DREN SUMINISTRADA POR: BIMSA						
17		DISTRIB. DE DRENAJES: (C/ MULTIPLE SI ES BOMBA BB)						
18		VALVULA DE VENTEO SUMINIST. POR: BIMSA						
19		DISTRIBUIDOR DE VENTEOS:						
20		CONST. ROSCADA. PARA SERV. DE TUBERIA <= 50°C: (6.4.3.2)						
21		ACCESORIOS ESPECIALES PARA TRANSICION: (6.4.3.3)						
22		REQUIERE ROSCAS CILINDRICAS: (6.4.3.8) NO						
23		REQUIERE SOPORTE DE REFUERZO: (6.4.3.10) NO						
24		CONEXIONES MAQUINADAS Y ATORNILLADAS: (6.4.3.12)						
25		TIPO DE CONEXIÓN DE DREN VS6: (9.3.13.5)						
26		DREN PARA EL EXTREMO DEL PATIN: NO						
27								
28		MATERIAL (6.12.1.1)						
29		CLASE DE MATERIAL ANEXO H: S-6						
30		TEMPERATURA DE DISEÑO MINIMA DEL METAL: (6.12.4.1) °C						
31		REDUCCION DE DUREZA DE MATERIALES: (6.12.1.12.1)						
32		ESTANDAR DE DUREZA APLICABLE: (6.12.1.12.1)						
33		CARCASA: ASTM A-516 / A-106						
34		BARRIL: ASTM A-516 / A-106 / A-105						
35		DIFUSORES: ASTM A-216 GR. WCB						
36		IMPULSOR: ASTM A-487 GR. CA6NM						
37		ANILLO-DESG. CARCAS./IMPUL.: ASTM A-479 TIPO 410 / ASTM A-276 TIPO 410						
38		FLECHA: ASTM A-276 TIPO 410						
39		TAZON (SI ES TIPO VS): ASTM A-216 GR. WCB						
40		CLASE DE INSPECCION: (API/ISOTABLA 14)(NOM-18-STPS-2000)						
41								
42		COJINETES Y LUBRICACION (6.10.1)						
43		COJINETES (TIPO / NUMERO):						
44								
45		RADIAL: CHUMACERAS DE COLUMNA LUBRICADAS POR EL MISMO LÍQUIDO						
46		EMPUJE: EN MOTOR ELÉCTRICO LUBRICACIÓN GRASA						
47		REVISION Y APROB. TAMAÑO COJINETES EMPUJE: (9.2.5.2.4)						
48		TIPO DE LUBRICACION: (6.11.3)(6.11.4)(9.2.6.1)						
49		SIST. DE LUBRICACION PRES. ISO10438- (9.2.6.4)						
50		HOJAS DE DATOS ADJUNTAS DE ISO 10438						
51		SISTEMA DE LUBRIC. PRESURIZADO MONTADO EN LA BASE:						
52		LOCALIZACION DEL SIST. LUBRIC. PRESURIZADO MONTADO EN LA BASE:						
53								
54		SUMIN. DE TUBERIA DE INTERCONEXION:						
55		VISCOSIDAD DEL ACEITE GRADO ISO VG:						
56		ACEITERA DE NIVEL CONSTANTE: (6.10.2.2)						
57		NOTAS						
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No. HOJA DE DATOS:	BIM-HDT-TP P55	Rev	0	HOJA	3	de	10
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HOJAS DE DATOS DE BOMBA
CENTRIFUGA

Ciente: *
No. de H.D.: BIM-HDT-TP P55
No. de serie: AL FABRICAR
No. REQ / ESPC: *
No. O.C.: *
Rev: 0 Hoja: 4 de 10

1	Nota	INSTRUMENTACION	MONTAJE DEL SISTEMA DE SOPORTE DE SELLOS	REV
2		VER HOJA DE DATOS ADJUNTA API-670: NO	SIST. SOPORT. DE SELLOS MONT. EN BASE DE BOMBA:(7.5.1.4) SI	
3		ACELEROMETRO O VELOMETRO: (7.4.2.1)	INDENTIF. DE LA LOCALIZA. EN BASE: (EN BASE CUANDO ESTE EQUIP. C/ DOB.SELLO)	
4		CANTIDAD:		
5		LOCALIZAC. DE MONTAJE:	TUB. DE INTERCONEX. POR: (PROV. DE BOMBA SI ES DOBLE SELLO) PROVEEDOR	
6			ENVIAR DEPOSITO POR SEPARADO : (NO SI ES DOBLE SELLO) NO	
7		PROVI. ROSCADAS SOLO POR ANSI/API 670: (6.10.2.10)		
8		CANTIDAD:		
9		LOCALIZAC. DE MONTAJE:	SELLO MECANICO (6.8)	
10			VER HOJA DE DATOS ADJUNTA ISO 21049/API 682: SI	
11		PARA MAGNETICA SE REQ. SUPERF. PLANA P/U's: (6.10.2.11)	PUERTO DE LIMPIEZA CENTRAL ADICIONAL: (6.8.9)	
12		CANTIDAD:	REQ. CHAQUETA DE CALENT. O ENFRI.: N/A	
13		LOCALIZAC. DE MONTAJE:		
14			ENFRIAMIENTO Y CALENTAMIENTO	
15			REQ. ENFRIAMIENTO: (6.1.17) SUMINIS. SI ES REQ. POR LA BOMBA	
16		OBETAS DE VIBRACION PARA LOS RODAMIENTOS HIDRODINAMICC (≥ 1492 kW)	PLAN DE TUBERIA PARA AGUA DE ENFRIAMIENTO:	
17		SUMINISTRO DE PROBETAS DE VIBRACION: (7.4.2.2)	TUBERIA PARA AGUA DE ENF.: TIPO DE ACCESORIOS:	
18		CANTIDAD POR COJINETES RADIALES:	MAT. DE LA TUBERIA DEL AGUA DE ENF.	
19		CANTIDAD POR COJINETES AXIALES:	MAT. DE LA TUBERIA DE AGUA DE ENF.:	
20		VIBR. MONITORS & CABLES SUPPLIED BY: (7.4.2.4)	CAJA DE BALEROS: l/m	
21			PLANES. AUX. AL SELLO: (INTERC,AMORT,BARRER,ETC.) l/m	
22		MEDIDORES DE TEMP. PARA RODAMIENTOS HIDRODINAMICOS: (7.4.2.3)	AGUA DE ENFRIAMIENTO TOTAL: l/m	
23		PROVISION-SOLO PARA MEDIDORES DE TEMPERATURA:	MEDIO DE CALENTAMIENTO:	
24		DETECTOR DE TEMPERATURA DE COJINETES RADIALES:	OTRO:	
25		CANTIDAD POR COJINETE RADIAL:	TUBERIA DE CALENTAMIENTO:	
26		DETECTOR DE TEMPERATURA DE COJINETES EMPUJE:		
27			TUBERIA Y ACCESORIOS	
28		CANTIDAD POR COJINETE DE EMPUJE LADO ACTIVO:	SIST. DE ARREGLO DE TUBERIA DE CONEXION. POR EL COMPRADOR : (7.5.1.6)	
29		CANTIDAD POR COJINETE DE EMPUJE LADO INACTIVO:	VENTEOS: SI	
30		MEDIDORES DE TEMPERATURA (CON TERMOWELL): (9.1.3.6)	DRENES: SI	
31		TIPO MEDIDOR DE PRESION:	AGUA DE ENFRIAMIENTO:	
32			TAGS EN TODOS LOS ORIFICIOS: (7.5.2.4) SI	
33			CONEXION SOLDADA EN SELLO DE ESTOPEROS: (7.5.2.8) (VER H.4) SI	
34				
35				
36		NOTAS		
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		HOJAS DE DATOS DE BOMBA CENTRIFUGA					Cliente: * No. de H.D.: BIM-HDT-TP P55 No. de serie: AL FABRICAR No. REQ / ESPC: * No. O.C.: * Rev: 0 Hoja: 5 de 10		
1	Nota	PREPARACION DE LA SUPERFICIE Y PINTURA					PRUEBA E INSPECCION		REV
2		ESTANDARES DEL FABRICANTE: NO					INSPECCION EN FABRICA: (8.1.1)		SI
3		OTRO (VER ABAJO): SI					CURVA DE COMPORTAMIENTO Y DAT. DE APROB. ANTES DE ENVIO:		SI
4		NO. DE ESPECIFICACION:					PRUEBA CON SELLO SUSTITUTO: (8.3.3.2.b)		
5		BOMBA:					REQ. CERTIF. DE MATERIAL: (6.12.1.8) CARCASA:		SI
6		PREPAR. DE SUPERF. DE BOMBA: SAND BLAST					(REF. TODAS LAS PART. MOJADAS, INC. LA TUB.) IMPULSOR:		SI
7		PRIMER / ENLACE EPOXICO: RP-4B					FLECHA:		SI
8		RECUBRIMIENTO FINAL: RA-26 + RA-28					OTRO:		
9		BASE:					REQ. PROCED. DE APROB. DE REPA. DE MOLDE: (6.12.2.5)(6.12.3.1)		
10		PREPARACION DE SUPERFICIE: SAND BLAST					REQ. INSPECCION PARA CONEXION DE SOLDADURAS: (6.12.3.4.d)(6.12.3.4.e)		
11		PRIMER / ENLACE EPOXICO: RP-4B					PARTICULAS MAGNETICAS:		
12		RECUBRIMIENTO FINAL: RA-26 + RA-28					RADIOGRAFIA:		
13							LIQUIDOS PENETRANTES:		SI
14		DETALLES DISPOSITIVO DE IZAJE: APLICA					ULTRASONIDO:		
15		BARQUE: (8.4.1)					INSPECCION REQUERIDA PARA CARCASAS:		
16		REQUIERE CONTENEDOR DE EXPORTACION: NO					PARTICULAS MAGNETICAS:		SI
17		ALMACENAJE EXTERIOR POR MAS DE 6 MESES: SI					RADIOGRAFIA:		
18		EMPAQUE DEL ROTOR DE REPUESTO PARA:					LIQUIDOS PENETRANTES:		
19		ORIENTACION DEL ALMACENAJE DEL ROTOR: (9.2.8.2)					ULTRASONIDO:		
20		ENVIO Y CONTENEDOR PARA ALMACENAJE VERTICAL.: (9.2.8.3)					REQ. PRUEBA DE DUREZA: (8.2.2.7)		
21		PURGA N2: (9.2.8.4)					EXAMEN SUB-SUPERFICIAL ADICIONAL: (6.12.1.5)(8.2.1.3)		
22		PARTES DE REPUESTO:					PARA:		
23		ARRANQUE: SI					METODO:		
24		MANTENIMIENTO NORMAL: SI					PRUEBA PMI REQUERIDA: (8.2.2.8)		
25							COMPONENTES P/SER EXAM.: PIEZAS EN CONTACTO		
26		PESOS KG					PRUEBA DE DESBALANCEO RESIDUAL: (J.4.1.2)		
27		TAG N°	BOMBA	ACCIONADOR	ENGRANAJE	CAB. DESCARGA	TOTAL	NOTIF. DE PRUEB.DE COMPORTAMIENTO DE FABRICA: (8.1.1.c)(8.3.3.5)	
28		TP P55	522	96		300	918	PRUEBA DE LA BASE: (7.3.21)	
29								PRUEB. HIDROSTATICA DE LA CARCASA:	NO ATEST
30								PRUEBA HIDRO. EN TAZONES Y COLUMNA: (9.3.13.2)	
31							PRUEBA DE COMPORTAMIENTO:		ATEST
32		OTROS REQUERIMIENTOS DEL COMPRADOR					PRUEBA EN CUMPLIMIENTO CON: (8.3.3.2)		
33		REQUIERE JUNTA DE COORDINACION: (10.1.3) SI					PUNTOS DE DATOS DE PRUEBA PARA: (8.3.3.3)		
34		PRESION MAXIMA DE DESCARGA PARA INCLUIR:					TOLERANCIA DE PRUEBA PARA : (8.3.3.4)		
35		DENSIDAD RELATIVA MAXIMA:					PRUEBA NPSH (8.3.4.3.1)(8.3.4.3.4) WHEN MARGIN < 10% OR 1M		
36		OPERACION PARA VELOCIDAD DE DISPARO:					PRUEBA NPSH-PA 1a. ETAPA UNICAMENTE: (8.3.4.3.2)		
37		DIA. MAX. IMPULSORES Y/O NO. DE ETAPAS:					PRUEBAS NPSH DE ACUERDO HI 1.6 O ISO 99062/5/2013: (8.3.4.3.3)		
38		APROBACION DEL DISEÑO DE CONEXIONES: (9.2.1.4)					NPSHA DE PRUEBA LIM. AL 110% DE NPSHA DE SITIO: (8.3.3.6)		
39		INFORME DE ANALISIS TORSIONAL: (6.9.2.10)					REPETIR PRUEBA POR FUGA DE SELLO: (8.3.3.2.d)		
40		INFORMES DE PROGRESO: SI					REPETIR PBA. REQ. DESPUES DE LA CARGA FINAL ADJ.: (8.3.3.7.b)		
41		ESQUEMA DEL PROCED. DE PRUEBAS OPCIONALES: (10.2.5)					PRUEBA DE LA UNIDAD COMPLETA: (8.3.4.4.1)		
42		REQ. DATOS ADICIONALES PARA 20 AÑOS DE RETENCION: (8.2.1.1)					PBA. DE NIV. DE SONIDO:(8.3.4.5)		
43		ANALISIS LATERAL REQUERIDO: (9.1.3.4)(9.2.4.1.3)					LIMPIEZA ANTES DEL ENSAMBLE FINAL: (8.2.2.6)		
44		ANALISIS MODAL REQ. PARA BOMBAS VS : (9.3.9.2)					LOCALIZ. DE INSPECCION DE LINEAS DE LIMPIEZA:		
45		MONTAJE ROTOR DE BALANCE DINAMIC. POR ISO G1.0: (9.2.4.2.3)					PRUEBA DE CARGA EN LA TOBERA:		
46		LISTA DE INSTALACION EN PROPUESTA: (10.2.3.I)					REV. COPLANAR PARA MONTAJE DE SUPERFICIE AMORTIGUADAS:		
47		ANALISIS DE RESP. AMORTIGUADA DE ESTADO ESTAT. VFD: (6.9.2.3)					PBA. MECANICA DE FUNCIONAM. HASTA TEMP. DE ACEITE ESTAB.:		
48		RESPUESTA TORSIONAL TRANSITORIA: (6.9.2.4)					4 HR. PBA. MEC. DESPUES DE TEMP. ESTAB. DE ACEITE: (8.3.4.2.1)		
49		CALCULO DE VIDA DE CHUMACERA (6.10.1.1) & (6.10.1.6):					4 HR. PRUEBA MECANICA: (8.3.4.2.2)		
50		EVAL. DE RIESGO DE IGNICION POR EN 13463-1 POR EXPLOSION ATM: (7.2.15)					PRUEBA DE RESONANCIA DEL COJINETE: (8.3.4.7)		
51		DIBUJOS DE LA CARCASA PARA RETIRO DE GROSOR: (10.3.2.3)					PRUEBA DE RESONANCIA ESTRUCTURAL: (9.3.9.2)		
52		REQ. BRIDAS EN LUGAR DE UNIONES SOLDABLES: (7.5.2.8) SI					REMO. / INSPEC. DE COJINETES HIDROD. DESP. DE PBA.: (9.2.7.5)		
53		INCLUYE ESPECTROS DE VIBRACION GRAFICADOS: (6.9.3.3) SI					PRUEBA DE EQUIPO AUXILIAR: (8.3.4.6)		
54		PERNOS DE CONEXION: (7.5.1.7)					EQUIPO P/ SER INCLUIDO EN PRUEBA AUX.:		
55		PROHIBIDOS PERNOS RECUBIERTOS DE CADMIO:					LOCALIZA. DE PRUEBA P/ EQUIPO AUX.:		
56		PROVEEDOR PARA REPARAR Y HT RECORDS: (8.2.1.1.c)					PRUEBA DE IMPACTO: (6.12.4.3) POR EN 13445		
57		PROCED. DE PBA. ENVIADOS POR EL PROVEEDOR: (8.3.1.1)					POR ASME SECCION VIII		
58		ENVIO DE LA LISTA DE REV. EN LA INSPECCION: (8.1.5)					REMOVER CARCASA DESPUES DE LA PRUEBA:		
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62		NOTAS							
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No. HOJA DE DATOS: BIM-HDT-TP P55Rev: 0HOJA 5de 10

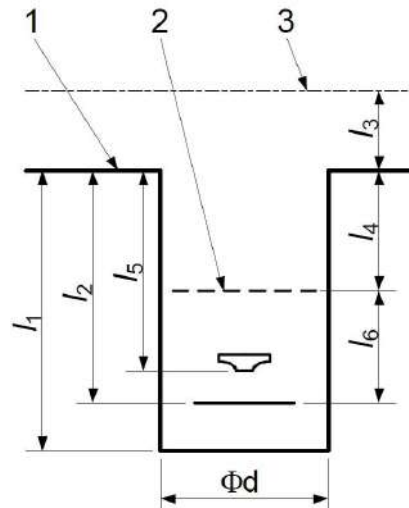
Cliente:		*	
No. de H.D.:		BIM-HDT-TP P55	
No. de serie:		AL FABRICAR	
No. REQ / ESPC:		*	
No. O.C.:		*	
Rev:	0	Hoja:	6 de 10

FOGV.01.03

HOJAS DE DATOS DE BOMBA CENTRIFUGA

Cliente: *
 No. de H.D.: BIM-HDT-TP P55
 No. de serie: AL FABRICAR
 No. REQ / ESPC: *
 No. O.C.: *
 Rev: 0 Hoja: 7 de 10

1	Nota	VERTICAL TIPO (FIG 1.1)	VS6	REV
2		OBSERV.		
3				
4				
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8		BOMBAS VERTICALES		BOMBAS VERTICALES (CONTINUACIÓN)
9		BOMBA DE EMPUJE:	(+) ARRIBA (-) ABAJO	FLECHA: CERRADO
10		EMPUJE ESTATICO:	N N	DIAMETRO DE LA FLECHA: 31.75 mm
11		MINIMO FLUJO:	1778 N N	DIAMETRO DE LA TUBERIA: mm
12		FLUJO NOMINAL:	1617 N N	ACOPLAMIENTO DE LA FLECHA:
13		MAXIMO FLUJO:	1093 N N	CONEXIÓN DE LA FLECHA:
14		MAXIMO EMPUJE:	1819 N N	- TIPO DE FILTRO DE ASPIRACIÓN: N/A
15		REQUIERE TAPA DE BASE:	NO	- NIVEL DE CONTROL:
16		TAPA DE BASE (LARGO X ANCH mm x mm		PINZAS DEL IMPULSOR ACEPTABLE: N/A
17		ESPESOR DE LA TAPA DE BAE: mm		MANGAS ENDURECIDAS BAJO CHUMACERA (9.3.10.5) N/A
18		SE REQUIERE MONTAJE DE BRIDA:		PRUEBA DE RESONANCIA: NO
19		COLUMNA TUBERIA:		ANALISIS ESTRUCTURAL(9.3.5): NO
20		DIAMETRO: mm		TORNILLOS DE ALINEACION DEL ACCIONADOR: NO
21		LONGITUD: mm		SUCC. DE LA LATA: LONGITUD: 3128.722 mm DIA.: 323.85 mm
22		NUMERO:		(VS6 & VS7) ESPESOR: mm
23		ESPACIO mm		PROFUNDIDAD DEL CARCAMO (VS1 HASTA VS5): N/A mm
24		BUJE GUIA:		SUMERGENCIA MINIMA (VS1 THROUGH VS5): N/A mm
25		NUMERO:		PLACAS DE MONTAJE SEPARADO (9.3.8.3.1): N/A
26		ESPACIO DEL COJINETE DE LA FLECHA mm		PLATO DE SEPARACION DEL MONTAJE (9.3.8.3.3): N/A
27		LUBRICACION DEL BUJE GUIA: ACEITE		TUBERIA DE DRENADO A SUPERFICIE N/A
28				REQUIERE CALCULO DE LA CARGA EN EL TAZON:
29		MATERIALES (adicional)		
30		CABEZAL DE DESCARGA:	ASTM A-516 / A-106 / A-105	MANGAS DE FLECHA:
31		TAZON:	ASTM A-216 GR. WCB	RETENEDOR DEL COJINETE:
32		FLECHA DE TAZON:	ASTM A-276 TIPO 410	TUB. QUE CUBRE LA FLECHA:
33		FLECHA:	ASTM A-276 TIPO 410	COLUMNA DE DESCARGA:
34		FLECHA CON CARA ENDUR.		ASTM A-516 / A-106 / A-105
35		CAMPANA EN SUCCION:		CAPACIDAD DE PRESION kg/cm²-g:
36		COJINETE DEL TAZON		MAWP HYDRO
37		COJINETE DE FLECHA		CARGA: 19.33 29.88
38				COLUMNA: 19.33 29.88
39				TAZON: 19.33 29.88
40		ARREGLO DEL SUMIDERO		
41				
42				
43				
44		DIMENSIONES DEL SUMIDERO:		
45		GRADO DE ELEVACIÓN	1 m	
46		NIVEL BAJO DEL LÍQUIDO	2 N/A m	
47		L.C. DE DESCARGA	3 0.23 m	
48		PROFUNDIDAD DEL SUMIDERO	l ₁ 3.13 m	
49		LONGITUD DE LA BOMBA	l ₂ 2.99 m	
50		GRADO A LA DESCARGA	l ₃ N/A m	
51		GRADO A LÍQUIDO BAJO LVL	l ₄ N/A m	
52		GRADO A 1A. ETAPA DEL IMPULSOR	l ₅ 2.67 m	
53		SUMERGENCIA REQUERIDA	l ₆ N/A m	
54		DIÁMETRO DE LA LATA	Φd 0.32 m	IN
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FOGV.01.03

HOJA DE DATOS DEL SELLO MECÁNICO

Cliente: *
No. de H.D.: BIM-HDT-TP P55
No. de serie: AL FABRICAR
No. REQ / ESPC: *
No. O.C.: *
Rev: 0 **Hoja:** 8 **de** 10

1	Nota	Cliente: * Servicio: BOMBAS PARA MANEJO DE CONDENSADO No. Requerido: 2 (DOS)	Proveedor: BIMS Planta: * Sitio: *	REV
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		HOJA DE DATOS DEL SELLO MECÁNICO		Cliente: * No. de H.D.: BIM-HDT-TP P55 No. de serie: AL FABRICAR No. REQ / ESPC: * No. O.C.: * Rev: 0 Hoja: 9 de 10		
1	Nota	Datos del Fluido (retenedor, barrera, amortiguador)		Sistema Auxiliar del Sello (Lineas de Conexión, Componentes, Instrumentación)		REV
2		Fluido Retenedor (plan 51, 62)		General		Reservorios Barrera y Amortiguador (planes 52, 53A)
3		☒ Tipo o Nombre		☐ Diseño Comprador/Proveedor (8.1.3)		☐ Proveedor/código ref.
4		☒ Temperatura Req. Máx °C		☐ Líneas de conexión aux.		☒ Dimensiones Estándar
5		Min °C		☐ soporte		☒ (8.3.6.2.7) Alternativo
6		☐ Flujo requerido Máx l/min		☐ Límites de Diseño para la Presión de la Cárcasa (bomba)		☒ Otro
7		Min l/min		MAWP kg/cm²-g		☒ Capacidad 12L d(shaft) ≤ 60 mm
8		Fluido Amortiguador/Barrera (Líquido/gas)		Máx. Temp. Permitida °C (8.1.4)		(8.3.6.2.5 a) 20L d(shaft) > 60 mm
9		☒ Tipo o nombre		☒ Límites de Diseño Sistema Aux. del Sello		Otro L
10		☐ Selección: Comprador		MAWP si >20/40 kg/cm²-g kg/cm²-g		☒ Altura - Glándula del sello / NLI m (8.3.6.2.2)
11		☐ Proveedor		Máx. Temp. Permitida °C		☐ ASME B31.3 (8.3.6.2.8)
12		☐ Revisión: Comprador		Comentario		☐ ASME Código de Stamp
13		☐ Proveedor		☐ Componentes Opcionales		Acumulador de Vejiga (Plan 53B)
14		☐ Flujo requerido Máx l/min		Válvula de alivio MAWP cárc. bom. >		☐ Proveedor/código ref.
15		Min l/min		En la succión de la bomba MAWP sistema aux.		☒ Tamaño Nominal 20 l d(shaft) ≤ 60 mm
16		☒ Calent./Enfriam. (+/-) kW		Válvulas de alivio térmicas (9.8.3)		35 l d(shaft) > 60 mm
17		☒ Presión de suministro (< camara del sello plan 52, 55, 72)		Circulación ext. Bomba (8.3.7) plan 23,52,53		Diseñado l (Tabla 9)
18		Máx kg/cm²-g		Conexión para Colector (8.2.24)		Acumulador de Pistón (Plan 53C)
19		Min kg/cm²-g		Mango de Restricción (8.3.5.4) línea lavado		☐ Proveedor/código ref.
20		☒ Temperatura de Normal °C (sólo líquido)		Otro, especifique		☒ Rango de Presión
21		Oper. del Fluido Máx. °C		☐ Requerimientos Especiales		☒ Volumen de Líquido Barrera (8.3.6.4.2)
22		Min. °C		Aleaciones Especiales		min/máx 0.2 / 2.8 l d(shaft) ≤ 60 mm
23		☒ Gravedad Especifica a Temp. Indicada		Servicio Peligroso		min/máx 0.35 / 5.1 l d(shaft) > 60 mm
24		@ Temp Máx kg/m3		Otro, especifique		Sistemas Externos de Lavado del Sello (plan 54 y 55)
25		@ Temp Normal kg/m3		☒ Periodo de Suspensión del Sist Días plan 52, 53		☐ Proveedor/código ref.
26		☒ Presión de vapor absoluta a Temp. Indicada		☒ Sistema de relleno		Reservorio para Fugas (Plan 65, 75)
27		@ Temp Máx kg/cm²-a		☒ Aislamiento para superficie calik		☐ Proveedor/código ref.
28		@ Temp Normal kg/cm²-a		Protección		Sist. Suministro de Gas Amortiguador/Barrera (Plans 72,74)
29		Pto. de Ebullición °C		☐ Roscas cónicas en tubería (8.2.13)		☐ Proveedor/código ref.
30		Viscosidad a Temp. Normal cP		ASME B1.20.1		☒ Válvulas Control Presión (9.9)
31		☒ Capacidad de Calor Especifico a Presión constante		ISO 7		☒ Amplificador de Presión (9.10)
32		del líquido a temp. Normal J/kg.K		Lineas de Conexión		INSTRUMENTACIÓN (Ver Anexo G)
33		SITIO Y SERVICIOS		☒ Tubing (8.2.8, Tabla 6)		☐ Temp. Diseño > 100 °C °C (9.1.7)
34		☐ Voltaje V		☐ Tubería (8.2.9, Tabla 5)		☐ IEC 604529, diseño IP65
35		☐ Frecuencia Hertz		☐ Catalogar		☐ NEMA 250, encl Tipo 4 (9.1.2)
36		☐ Fases		☐ Tamaño 12 mm X d(shaft) ≤ 60 mm		Unidades (presión,temp., nivel, flujo)
37		☐ Area Eléctrica CL		20 mm d(shaft) > 60 mm		Indicador de Temp. Para plan
38		DIV		otro mm		☐ bi-metalico (9.2.1) Opción para tubería
39		GPO		☐ Rastreo de Calor (8.2.23)		☐ líquido lleno planes 32, 53B, 53C
40		☐ Temp. Ambiente Máx. °C (Diseño)		Enfriador (plan 21, 22, 23, 41, 53B, 53C)		Instrumentos de Presión
41		Min. °C		☐ Proveedor/código ref.		☐ Medidores para el plan
42		☐ Agua de Enfriamiento		☐ ASME B31.3 (8.3.2.1.2)		☐ Aceite Lleno (medidor) (9.4.3)
43		☐ Otro líquido de Enfriamiento		☒ Enfriador por Agua (8.3.2.2)		☐ Transmisor para plan
44		☐ Temperatura °C		☐ Tub. Galvanizada AC (8.2.21)		☐ Interruptor para plan
45		☒ Flujo l/min		☒ Corriente Natural de Aire (8.3.2.3.1)		☐ Configurar Presión mín. kg/cm²-g
46		☐ Contenido de Cl2 ml/m3		☒ Corriente Forzada de aire (8.3.2.3.2)		máx kg/cm²-g
47		☐ Presión Normal kg/cm²-g		Separador de Ciclón		Instrumentos de Nivel (plan 52, 53A, 75)
48		☐ Presión Diseño kg/cm²-g		☐ Proveedor/código ref.		☐ Indicador Soldado (9.6.1)
49		☐ ATEX Grupo				Removible (9.6.2)
50		Cat. EC Directive				☐ Transmisor Hidrostático (9.5.3)
51		Clase T 94/9/EC				Capacidad
52						Ultrasónico
53		PRUEBAS E INSPECCIÓN				☐ Interruptores (9.5.1.8)
54		Prueba al Equipo de Bombeo - Sello		Diseño de Soldadura		☐ Puntos de Ajuste Alto
55		☒ Caras del sello modificadas		☐ Aprobación del comprador (6.1.6.10.5)		Bajo
56		☒ Sello alternativo		☐ Método para 100% inspección a soldaduras (6.1.6.10.5)		Instrumentos de Flujo
57		☐ Miscelaneo		☐ Partículas Magnéticas		☐ Indicador de Flujo para Plan N/A
58		☐ Prueba de Calidad Optima (10.3.2.1.2)		☒ Líquidos Penetrantes		e.g. en línea de salida de Agua de Enfriamiento (8.2.22) o plan 32
59		☐ Prueba de Dureza por		☐ Radiografiado		☐ Medidor de Flujo para plan
60		☐ Participación del Comprador		☐ Ultrasonido		☐ Transmisor para Plan
61		☐ Checklist del Inspector (Anexo H)				☐ Interruptor para Plan
62		NOTAS				
63						
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FOGV.01.03						
No. HOJA DE DATOS:		BIM-HDT-TP P55		Rev: 0	HOJA 9 de 10	

HOJAS DE DATOS MOTOR ELÉCTRICO

Cliente: *
No. de H.D.: BIM-HDT-TP P55
No. de serie: AL FABRICAR
No. REQ / ESPC: *
No. O.C.: *
Rev: 0 **Hoja:** 10 **de** 10

No. de proyecto: VER PORTADA **Centro de trabajo:** _____
Clave del equipo: TP P55 **Planta del centro de trabajo:** _____
Carga a accionar: _____ **Cantidad de equipos:** _____

CONDICIONES AMBIENTALES.			
	Bulbo seco / Húmedo		
Temperatura máxima:	____ °C / ____ °C	Humedad relativa:	Min: ____ %
Temperatura mínima:	____ °C / ____ °C		Máx: ____ %
Temperatura promedio:	____ °C / ____ °C	Altitud de operación:	____ m.s.n.m.
Ambiente seco: ()	Ambiente húmedo: ()	Ambiente húmedo con salinidad y gases derivados del azufre: ()	Ambiente marino: ()

DATOS GENERALES			
SISTEMA ELÉCTRICO			
Tensión del sistema:	460 V	Fases:	3 Frecuencia: 60 Hz
DATOS DEL MOTOR			
Potencia nominal:	7.456/ 10 kW/hp	Tensión nominal:	460 V c.a. Letra de diseño: B Armazón No.: 215HP Eficiencia nominal: 91.7 %
Corriente nominal:	12.8 A	Número de fases:	3 Clase de aislamiento: F Sobrevelocidad: ____ % Factor de potencia: 79.5
Frecuencia eléctrica:	60 Hz	Frecuencia de rotación nominal:	1800 Frecuencia de rotación (sincronas): 1800 r/min No. Polos: 4
Inercia WK ² :	51 LB-FT ²	Eficiencia Premium:	91.7 % Factor de servicio: 1.15 Corriente de Arranque a Rotor Bloqueado y/o Letra de Código: 88 / H Eficiencia Energética: 91.7 %

ENVOLVENTE			
MOTOR TOTALMENTE CERRADO		TROPICALIZADO (X)	Grado de protección: IP55
TEAAC (TCEAA): ()	TEEP (TCPE): ()	TEFC (TCVE): (X)	TEIGF (TCPGI): ()
TEPV-IP (TCDV-IP): ()	Gasto: ____ m³/min Presión: ____ Kpa Tem. Max.: ____ k	TEWAC (TCEAAG): ()	Gasto: ____ m³/min Presión: ____ Kpa Tem. Max.: ____ k

MOTOR A PRUEBA DE EXPLOSIÓN			
Clase:	Clase I: (X) Clase II: ()	División:	División 1: () División 2: (X)
Nivel de ruido máximo: 65 dB(A).	Código de temperatura T: T3 Elevación de Temperatura: ____	Acabado: ____	Sistema 1: () Sistema 2: () Sistema 3: ()

TIPO DE ARRANQUE			
Tensión plena no reversible: (X)	Tensión plena reversible: ()	Tensión plena varias velocidades: ()	Arranque Suave: () Variador de frecuencia: ()
Otro: () Indicar: _____			
Motor uso variador de frecuencia	Par constante: ()	Par variable: ()	Relación: ____:1

MONTAJE			
Horizontal: () Vertical: (X)	Con base: () Sin base: ()	Brida: Si: () No: ()	Tipo: C: () D: () Otras: _____
Para motor vertical: Flecha Sólida: (X) Flecha Hueca: ()	Empuje Normal: ()	Empuje Alto: (X)	Flecha especial especificar: ()
Flecha especial Material: _____	Diámetro: _____	Longitud: _____	Trinquete de no retocoso: SI () NO ()
Rotación (vista desde lado cople) CW: (X) CCW: () Ambos: ()	Acoplamiento Directo: (X)	Bandas: ()	Cadenas: ()
Rodamiento lado cople:	Fabricante: _____ Tipo/modelo: BOLAS 6309-J/C3	Tamaño: _____	
	Tipo de lubricación: GRASA	Tipo de lubricante: _____	SAE No.: _____
Rodamiento opuesto al cople:	Fabricante: _____ Tipo/modelo: BOLAS 6208-J/C3	Tamaño: _____	
	Tipo de lubricación: GRASA	Tipo de lubricante: _____	SAE No.: _____
Lubricación por niebla: Si: () No: (X)			

ACCESORIOS			
Resistencias calefactoras: Si: () No: (X)	Tensión 120 V c.a.	Fases: 1	Frecuencia: 60 Hz
Detector de vibración radial: ()	Cantidad puntos: _____	Alarma: ()	Disparo: () Tipo: _____
Detector de vibración axial: ()	Cantidad puntos: _____	Alarma: ()	Disparo: () Tipo: _____
Detector temperatura de rodamientos: ()	Cantidad puntos: _____	Alarma: ()	Disparo: () Tipo: _____
Detector temperatura de devanados: ()	Cantidad puntos: _____	Alarma: ()	Disparo: () Tipo: _____

CAJA DE CONEXIONES			
Tamaño: _____	Normal: ()	Sobredimensionada: ()	Dimensiones: () Conos de alivio: () Detectores de temperatura: ()
Para alojar tubería conduit.	Cantidad: _____	Tamaño: _____	Devanados: () Rodamientos: ()

PRUEBAS			
Pruebas de rutina:	Atestiguada ()	No Atestiguada (X)	
Pruebas de diseño o de prototipo:	Atestiguada ()	No Atestiguada ()	
Pruebas complementarias:	Atestiguada ()	No Atestiguada ()	
Pruebas de campo:	Atestiguada ()	No Atestiguada ()	
Reporte de pruebas:			
Diagramas de conexiones de fuerza y control:	Si ()	No ()	Dibujos de dimensiones para: Aprobación: () Información: ()

NOTAS

BOMBA CENTRIFUGA VERTICAL DE LATA

CLAVE(S): TP P55

Servicio: BOMBAS PARA MANEJO DE CONDENSADO

Tipo: VS6 DEL API-610 12ª EDICIÓN

Modelo: 1305L

Tamaño: 8 FMV-501

No. Pasos: 3 (TRES)

Eficiencia: 71 %

Potencia al Freno en el punto de Operación: 4.78 HP

Potencia al Freno Máxima: 5.56 HP

Diámetro de Diseño del Impulsor: 147.6375 mm (5.8125").

Diámetro Máximo del Impulsor: 147.6 mm (5.813").

Diámetro Mínimo del Impulsor: 130.2 mm (5.125").

Brida de Succión: 8" Ø 150#, R.F. HORIZONTAL

Brida de Descarga: 6" Ø 150#, R.F. HORIZONTAL

Presión Máxima de Trabajo Permisible: 275 PSIG

Presión de Prueba Hidrostática: 425 PSIG

Tipo de Impulsor: CERRADO DE SIMPLE SUCCION

Balero Radial: DE CHUMACERAS DE COLUMNA LUBRICADAS POR EL MISMO LÍQUIDO

Balero Axial: DE BOLAS EN MOTOR ELÉCTRICO LUBRICACIÓN GRASA

Sello Mecánico: BIMSA o EAGLE BURGMANN

Tipo: API 682 4a EDICIÓN

CÓDIGO: BSTHN

Planes de lubricación al sello: 13 + 61

Material de los Planes: TUBING DE ACERO INOXIDABLE 316L

Acoplamiento Bomba – Accionador: DIRECTO MEDIANTE COPLER RIGIDO CON ESPACIADOR, MARCA BIMSA

Base Bomba – Accionador: ACERO ESTRUCTURAL ASTM A-36

Material de Construcción de acuerdo a COLUMNA S-6 DEL API-610 12ª EDICIÓN.

Tazón: ASTM A-216 GR. WCB

Impulsor: ASTM A-487 GR. CA6NM

Anillos de desgaste Impulsor / Tazón: ASTM A-276 TIPO 410 / ASTM A-479 TIPO 410

Flecha: ASTM A-276 TIPO 410

Cabezal de Descarga: ASTM A-516 / A-106 / A-105

Lata: ASTM A-516 / A-106

Guardacople: ALUMINIO

ACCIONADOR

**Tipo: MOTOR ELECTRIC VERTICAL DE INDUCCION JAULA DE ARDILLA FLECHA SOLIDA
CON TRINQUETE DE NO RETROCESO**

Marca: US

Potencia: 10 HP

Velocidad: 1800 RPM

Número de Polos: 4

Armazón: 215HP

CARCASA: TEFC

Voltaje / Fases / Hertz: 460 / 3 / 60

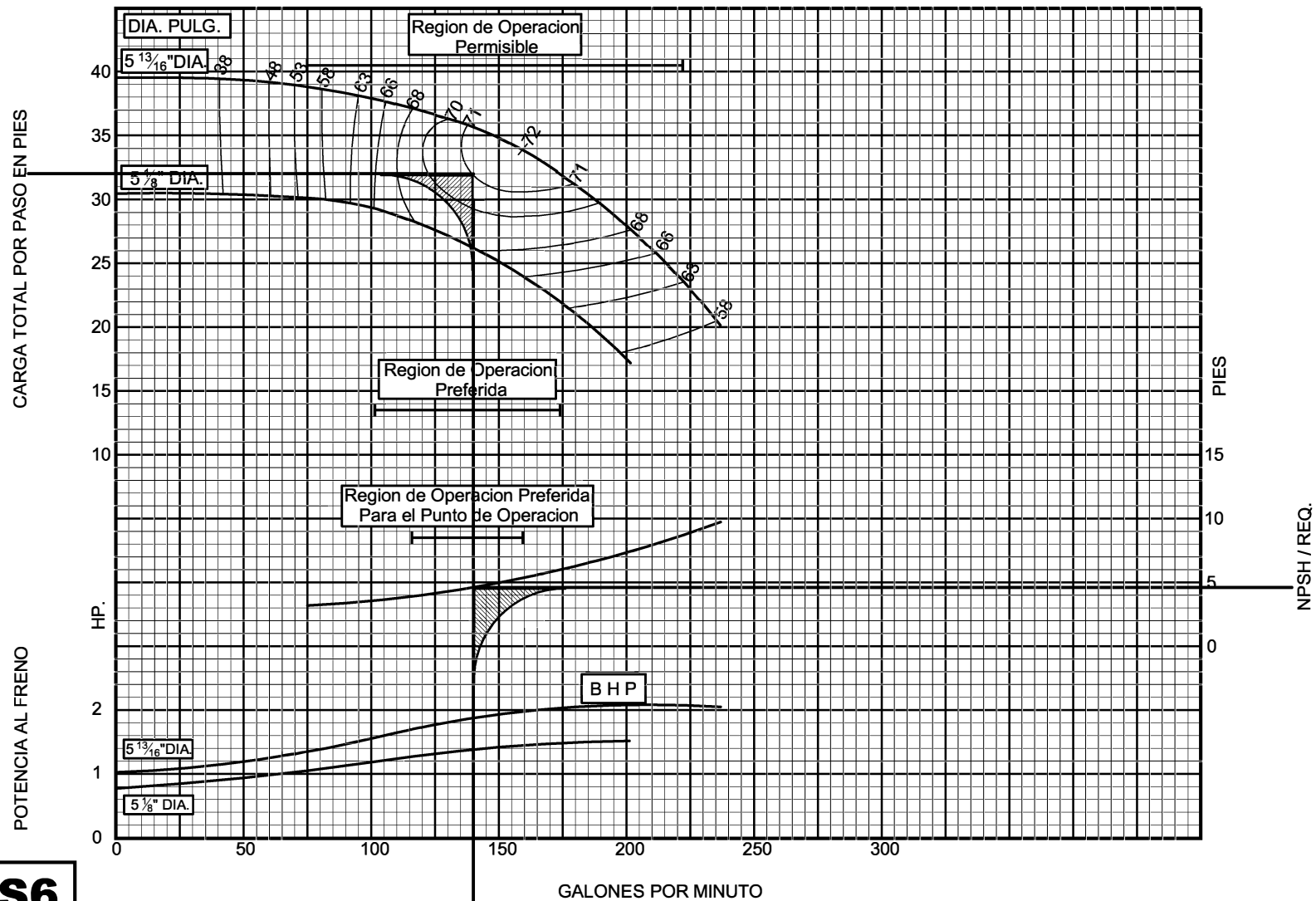
Tipo de Rodamientos: BOLAS

Lubricación: GRASA

Aislamiento: F TROPICALIZADO

Factor de Servicio: 1.15

Eficiencia: PREMIUM



TIPO VS6

MODELO. 1 3 0 5	RPM 1 7 6 0	TAMAÑO DE BOMBA: 8 FMV - 501 / 3 PASOS	No. CURVA: 8 x 6 - CB	BEP EN IMPULSOR DE SELECCION 149 GPM (33.86 m³/h)
		REQUISICION:	CONDICIONES DE OPERACION	GASTO MINIMO CONTINUO ESTABLE 75 GPM (17.05 m³/h)
		CARGA: 29.26 m (96 Pies)	EFICIENCIA: 71 %	GASTO MINIMO CONTINUO TERMICO 52.5 GPM (11.93 m³/h)
		CAPACIDAD: 31.82 m³/h (140 GPM)	POTENCIA AL FRENO: NOM. 4.78 HP / MAX. 5.56 HP	DIAMETRO MAX. 5.813" (147.6 mm)
		LIQUIDO: CONDENSADO	NPSHR: 1.4 m (4.6 Pies)	DIAMETRO MIN. 5.125" (130.2 mm)
		GRAVEDAD ESPECIFICA: 1.0	NPSHD: 1.40 m (4.6 Pies) + 1.50 m (4.8 Pies) = 2.90 m (9.52 Pies)	DIAMETRO DISEÑO. 5 13/16" (147.63 mm)
		VISCOSIDAD:	TEMPERATURA:	Ut. 21 Ft/s
		PRESION DE SUCCION: 0.0 kg/cm², man.	MOTOR: T E F C DE: 10 HP 4 POLOS	Nss. 6 3 7 3
		PRESION DE DESCARGA: 2.92 kg/cm², man.	CLIENTE:	No. IMPULSOR. BY-150250A
				PASO DE ESFERA. 1/8"
				CLAVE. TP P55
				FECHA. JUNIO / 2026
				CANTIDAD. 01 (UNA)

SELECCIÓN DEL MOTOR ELÉCTRICO

MEMORIA DE CÁLCULO

TAMAÑO DE LA BOMBA: 8 FMV-501 / 3 PASOS
CLAVE(S): TP P55

CALCULO DEL BHP.

$$BHP_{NOM.} = \frac{Q \times H \times Gr.Esp.}{3960 \times \eta_{\beta}}$$

$$BHP_{NOM.} = \frac{140.00 \text{ GPM} \times 96.00 \text{ PIES} \times 1.000}{3960 \times 0.710} = 4.78 \text{ HP}$$

$$BHP_{MAX.} = \frac{220.38 \text{ GPM} \times 57.97 \text{ PIES} \times 1.000}{3960 \times 0.580} = 5.56 \text{ HP}$$

PARA LA SELECCIÓN DEL MOTOR, LA POTENCIA EN EL PUNTO DE OPERACIÓN DEBE MULTIPLICARSE POR EL FACTOR DE 1.25 QUE MARCA EL PARRAFO 7.1.5 DEL API-610 12ª EDICIÓN. CON ESTE VALOR SELECCIONAMOS LA LA POTENCIA DEL MOTOR ELECTRICO:

$$BHP_{MOTOR.} = 4.78 \text{ HP} \times 1.25 = 5.98 \text{ HP}$$

MOTOR DE 10 HP 4 POLOS CON F.S. DE 1.15
--

MEMORIA DE CALCULO EMPUJE AXIAL

TAMAÑO DE LA BOMBA: 8 FMV-501 / 3 PASOS

CLAVE(S): TP P55

EMPUJE AXIAL= EMPUJE HIDRAULICO + PESO DEL ROTOR + PESO DE LA FLECHA DE LINEA

EMPUJE HIDRAULICO = CARGA DINAMICA TOTAL x FACTOR DE EMPUJE x GRAVEDAD ESPECIFICA

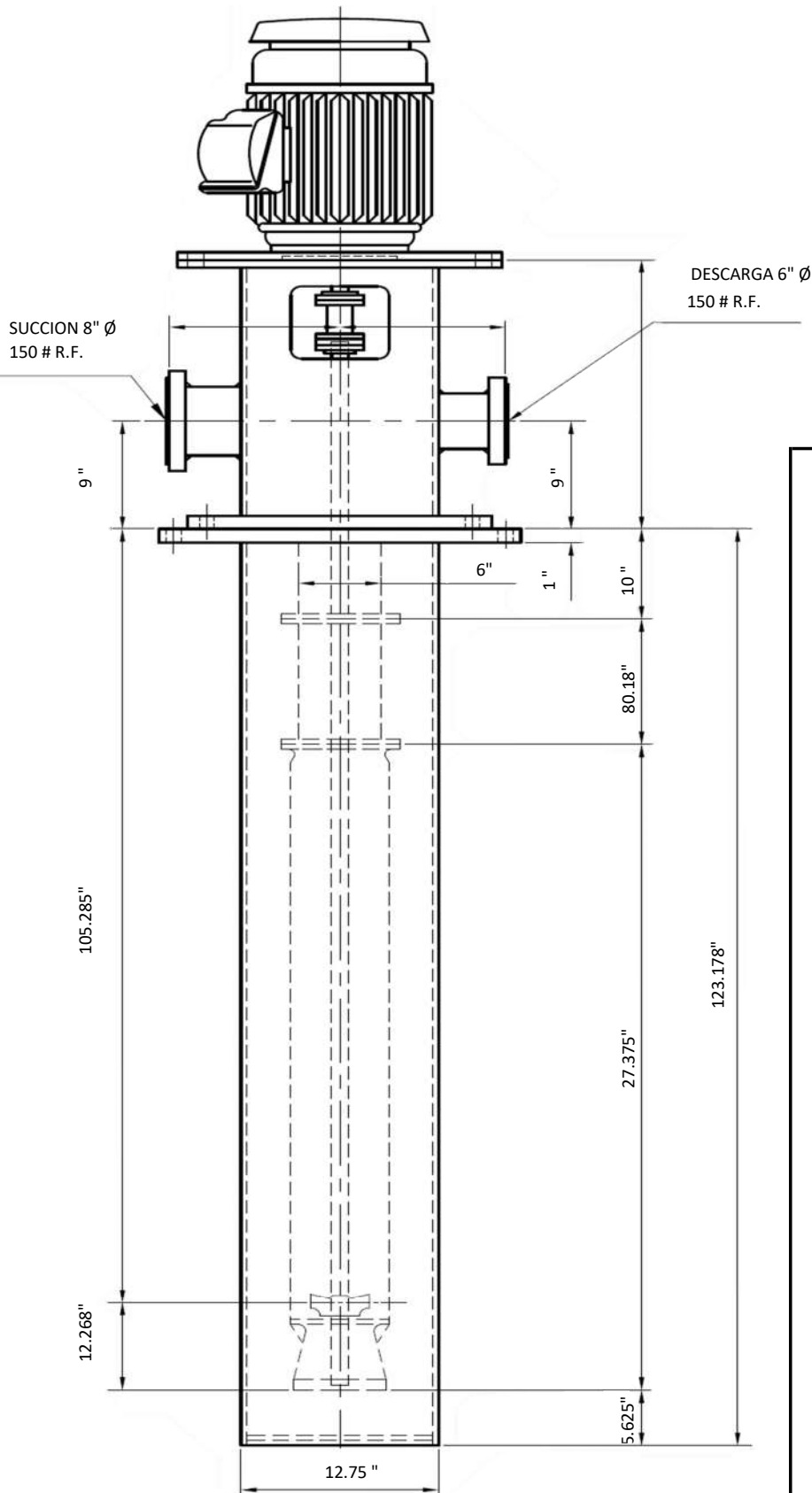
FACTOR DE EMPUJE= 3.1
GRAVEDAD ESPECIFICA= 1

No. DE PASOS DE LA BOMBA= 3 PASOS
PESO DEL ROTOR POR PASO= 10 LB

DIAMETRO DE LA FLECHA DE LINEA= 1.25 PULGADAS
PESO DE LA FLECHA DE LINEA= 4.16 Lb/pie
LONGITUD DE LA FLECHA DE LINEA= 8.65 PIES

	VALVULA CERRRADA	GASTO MINIMO CONTINUO ESTABLE	GASTO DE OPERACIÓN	GASTO MAXIMO
<u>C.D.T. (Pies)</u>	110.61	107.66	96.00	57.97
<u>EMPUJE HIDRAULICO (Lb)</u>	342.89	333.75	297.60	179.71
<u>PESO DEL ROTOR (Lb)</u>	30.00	30.00	30.00	30.00
<u>PESO DE LA FLECHA DE LINEA (Lb)</u>	36.00	36.00	36.00	36.00
<u>EMPUJE AXIAL (Lb)</u>	<u>408.89</u>	<u>399.76</u>	<u>363.60</u>	<u>245.71</u>

BOMBA CENTRIFUGA VERTICAL DE LATA
TAMAÑO 8 FMV-501 / 3 PASOS
MODELO 1305L



ACOTACIONES EN PULG.

DIBUJO SIN ESCALA

CLIENTE:

REFERENCIA:

TP P55

COLUMNA LUBRICADA POR:

POR EL MISMO LIQUIDO

DIAM. EXT. DE TAZON:

7.5"

DIAM. EXT. DE LATA:

12.75 "

CONDICIONES DE OPERACIÓN

CAPACIDAD:

140 GPM

CARGA TOTAL:

96 PIES

MATERIALES DE CONTRUCCION

de acuerdo a COLUMNA S-6 DEL API-610 12ª EDICIÓN

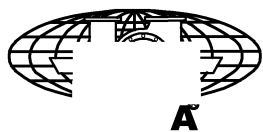
OBSERVACIONES:

LAS DIMENSIONES PUEDEN VARIAR

A SOLICITUD DEL CLIENTE

DIBUJO PRELIMINAR.

MOTOR: 10 HP , 4 POLOS

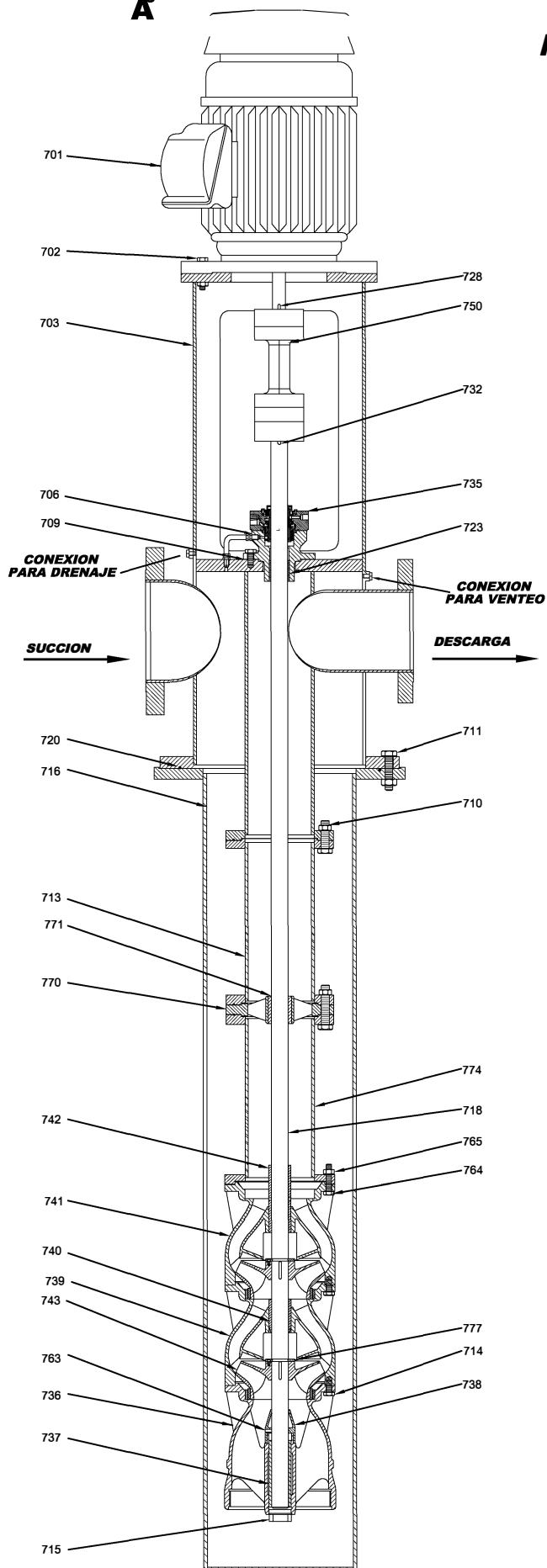


BOMBA CENTRIFUGA VERTICAL DE LATA

TIPO VS6 DEL API-610 12a. EDICION

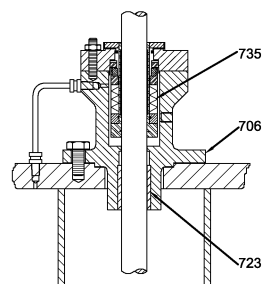
MODELO 1305L

CORTE SECCIONAL

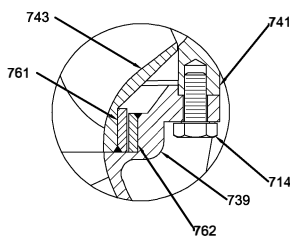


LISTA DE PARTES:

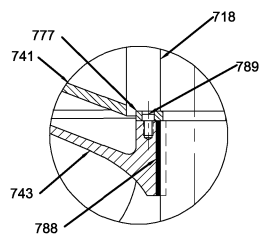
REF.	DESCRIPCION:
701	MOTOR ELECTRICO
702	TORNILLOS DE LA BASE DEL MOTOR
703	CABEZAL DE DESCARGA TIPO "T"
706	CAJA DE SELLO MECANICO
709	JUNTA DE LA CAJA DEL SELLO MECANICO
710	TORNILLOS DEL TUBO DE COLUMNA
711	TORNILLOS DEL CABEZAL
713	TUBO DE COLUMNA BRIDADO (SUPERIOR)
714	TORNILLOS DE LOS TAZONES
715	TAPON CONO DE ENTRADA
716	LATA
718	FLECHA DE IMPULSORES
720	ANILLO "O"-RING DEL CABEZAL DE DESCARGA
723	BUJE DE LA CAJA DEL SELLO MECANICO
728	CUÑA RECTANGULAR DE LA FLACHA DEL MOTOR
732	CUÑA RECTANGULAR DE LA FLACHA DE IMPULSOR
735	SELLO MECANICO
736	CONO DE ENTRADA
737	BUJE DEL CONO DE ENTRADA
738	CAPACETE DEL CONO DE ENTRADA
739	TAZON INTERMEDIO
740	BUJE DEL TAZON INTERMEDIO
741	TAZON SUPERIOR
742	BUJE DEL TAZON SUPERIOR
743	IMPULSOR
750	COPLE ESPACIADOR
761	ANILLO DE DESGASTE DE IMPULSOR
762	ANILLO DE DESGASTE DE TAZON INTERMEDIO, SUPERIOR Y CONO DE ENTRADA
763	PRISIONERO CAPACETE CONO DE ENTRADA
764	TORNILLO CABEZA HEXAGONAL (COLUMNA)
765	TUERCA
770	SOPORTE DE CHUMACERA
771	CHUMACERA
774	TUBO DE COLUMNA INFERIOR
777	ANILLO DE SUJESION DE IMPULSOR
788	CUÑA DE IMPULSOR
789	TORNILLO DE ANILLO DE SUJESION



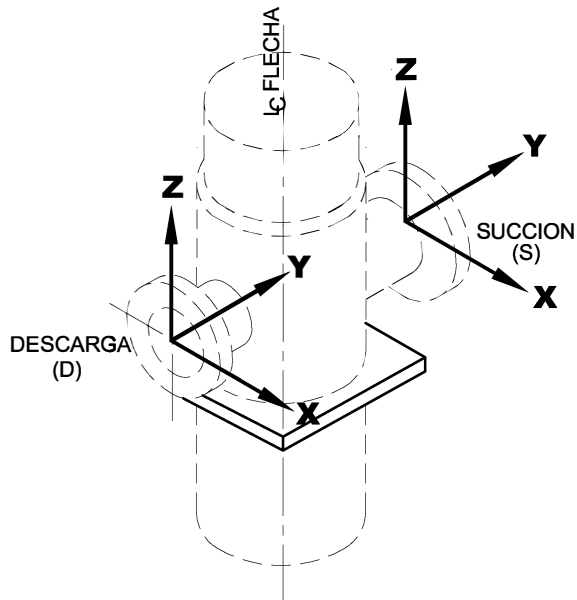
DETALLE DEL SELLO MECANICO



DETALLE DE ANILLOS DE DESGASTE



DETALLE DE LA SUJECION DE IMPULSOR



**SISTEMA DE COORDENADAS DE FUERZAS Y
MOMENTOS DE BOMBAS VERTICALES CON
SUCCION Y DESCARGA HORIZONTAL**

FUERZAS Y MOMENTOS	DIAMETRO NOMINAL DE BRIDA								
	≤ 2	3	4	6	8	10	12	14	16
DESCARGA HTAL									
FX	320	480	640	1120	1700	2400	3000	3200	3800
FY	400	600	800	1400	2200	3000	3600	4000	4600
FZ	260	400	520	920	1400	2000	2400	2600	3000
FR	580	860	1040	2020	3120	4400	5200	5800	6600

SUCCION HTAL.									
FX	320	480	640	1120	1700	2400	3000	3200	3800
FY	400	600	800	1400	2200	3000	3600	4000	4600
FZ	260	400	520	920	1400	2000	2400	2600	3000
FR	580	860	1040	2020	3120	4400	5200	5800	6600

MOMENTOS									
MX	680	1400	1960	3400	5200	7400	9000	9400	10800
MY	340	700	1000	1740	2600	3600	4400	4600	5400
MZ	520	1060	1480	1600	3800	5600	6800	7000	8000
MR	920	1900	2660	4620	7000	10000	12200	12600	14400

F = Fuerza en Libras
M = Momento en Libras-pie
R = Resultante

DIBUJO REALIZADO EN AUTO CAD

Características

API 682 Categoría 2, Tipo A,

Arreglo 2 sello

Sello Tandem

Balaceado

Cartucho

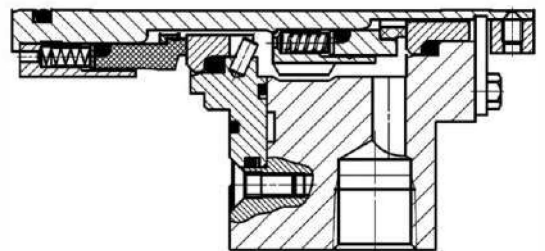
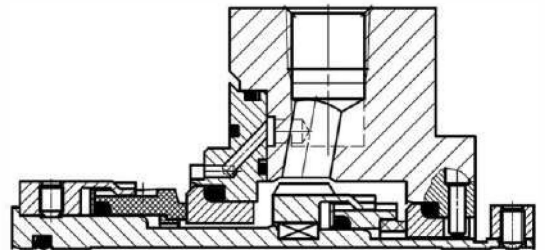
Cara sólida o insertada

Asiento sólido

Resortes protegidos del product

**Insensible a deflexiones del rotor y
fluctuaciones del proceso**

Conección Quench



LIMITES DE OPERACION

$d1 = 20 \dots 110 \text{ mm}$

$p1 = 42 \text{ barg}$

$t = -40 \dots 260 \text{ }^{\circ}\text{C}$

$vg = 20 \text{ m/s}$

Máximo Movimiento Axial Permitido:

$d1 \text{ hasta } 50 \text{ mm: } \pm 1.0 \text{ mm}$

$d1 \text{ } 50 \text{ hasta } 75 \text{ mm: } \pm 1.5 \text{ mm}$

$d1 \text{ desde } 75 \text{ mm hacia arriba: } \pm 2.0 \text{ mm}$

PLANES RECOMENDADOS

Lado proceso:

01, 02, 03, 11, 12, 13, 14*, 21, 22, 23*, 31, 32, 41

Entre sellos:

52, 55

Lado atmosférico:

51, 61, 62, 65A, 65B, 66A*, 66B*

*NO ESTANDAR

MATERIALES

Cara:

SiC (Q1, Q2)

**Asiento: SiC (Q1, Q2) Carbon A,
Carbon B**

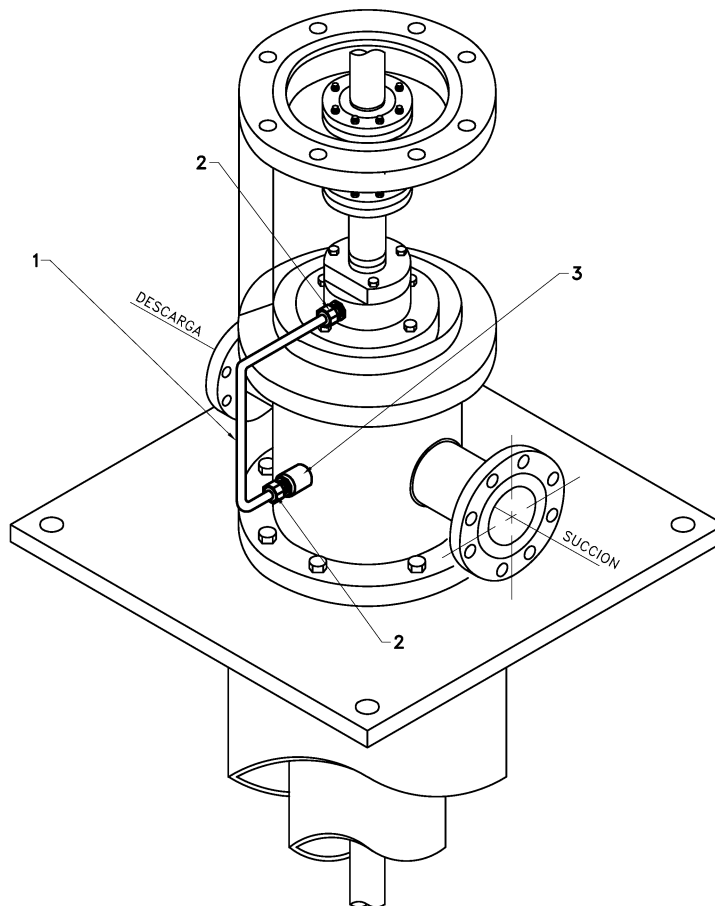
**Elementos de sellado secundario:
EPDM (E), FKM (V), FFKM (K)**

**Elementos de empuje :
Hastelloy C-4 (M)**

Partes metálicas:

**Acero CrNiMo tipo 316 (G)
o equivalente,**

Materiales opcionales por solicitud.



NOTA:
CONFIGURACION TIPICA SUJETA A CAMBIOS

3	ORIFICIO DE 1/2" Ø	1	A-479 TIPO 316L
2	CONECTOR MACHO 1/2"-1/2" Ø NPT	2	A-269 TIPO 316
1	TUBING 1/2" Ø, ESP. 0.065"	25'	A-269 TIPO 316
No.	DESCRIPCION	CANT.	MATERIAL



PLAN
13
API

RECIRCULACION DESDE LA CAJA DE SELLO A TRAVES DE
UN ORIFICIO Y RETORNO A LA SUCCION DE LA BOMBA

BOMBA : _____ TAMAÑO : _____
SERIE DEL EQUIPO : _____
CLIENTE : _____
USUARIO FINAL : _____
CLAVE (S) : _____
No. DE PEDIDO : _____
PARTIDA : _____ CANTIDAD : _____
REQUISICION : _____
PLANTA : _____
DESTINO : _____
CERTIFICADO POR : **ING. DANIEL MARTINEZ R.**
REVISO : **ING. ALFONSO MONROY M.** DIBUJO : **ING. ROGELIO GACHUZ G.**
No. DE DIBUJO : **B-90010-**

REVISIONES:

0.

RELY ON EXCELLENCE

H75VA4-S

API 682 4th edition product range | Mechanical seals | Balanced pusher seals



Features

- API 682 Category 2 and 3, Type A, Arrangement 1 seal
- Single seal
- Balanced
- Cartridge unit
- Bi-directional design
- Rotating multiple springs
- Integrated pumping device available

Advantages

- Compact design
- Universally applicable both for retrofits or original equipment
- Efficient stock-keeping due to standardized components
- Extended selection of materials
- Extended field of operation in terms of temperature and pressure
- Metal parts also in special materials available

Operating range

Shaft diameter:

$d_1 = 20 \dots 110 \text{ mm (0.79" ... 4.33")}$

Pressure: $p_1 = \dots 42 \text{ bar (609 PSI)}$

Temperature:

$t = -40 \text{ °C} \dots +176 \text{ °C (-40 °F} \dots +349 \text{ °F)}^*$

Sliding velocity: $v_g = 23 \text{ m/s (75 ft/s)}$

Axial movement:

$d < 50 \text{ mm} \pm 1,0 \text{ mm}$

$d \geq 50 \text{ mm} \pm 1,5 \text{ mm}$

* Engineered up to 260 °C (500 °F) with FFKM (K) secondary seals.

Materials

Seal ring: Blister resistant carbon, Silicon carbide SSiC (Q1), RBSiC (Q2)

Mating ring:

Silicon carbide SSiC (Q1), RBSiC (Q2)

Secondary seals:

EPDM (E), NBR (P), FKM (V), FFKM (K)

Springs: Hastelloy® C-276 (M5)

Metal parts: CrNiMo steel 316 (G) or equivalent, optional materials on request.

Standards and approvals

- API 682 / ISO 21049
- API 682 4th ed. Cat. 2/3 - 1CW-FL

Recommended applications

- Refining technology
- Oil and gas industry
- Petrochemical industry
- Chemical industry
- Power plant technology
- Highly volatile hydrocarbons
- LPG plants
- API 610 / ISO 13709 pumps
- Process pumps

Recommended piping plans

Process side:

[API Plan 01](#)

[API Plan 02](#)

[API Plan 03](#)

[API Plan 11](#)

[API Plan 12](#)

[API Plan 13](#)

[API Plan 14](#)

[API Plan 21](#)

[API Plan 22](#)

[API Plan 23](#)

[API Plan 31](#)

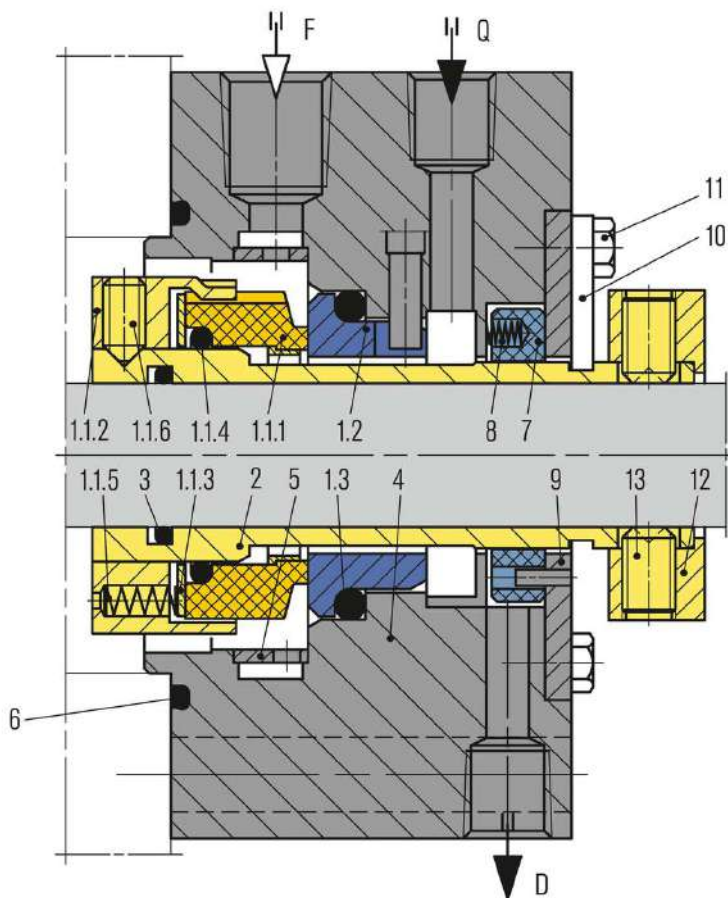
[API Plan 32](#)

[API Plan 41](#)

Atmospheric side:

RELY ON EXCELLENCE

API Plan 51
API Plan 61
API Plan 62
API Plan 65A
API Plan 65B
API Plan 66A
API Plan 66B

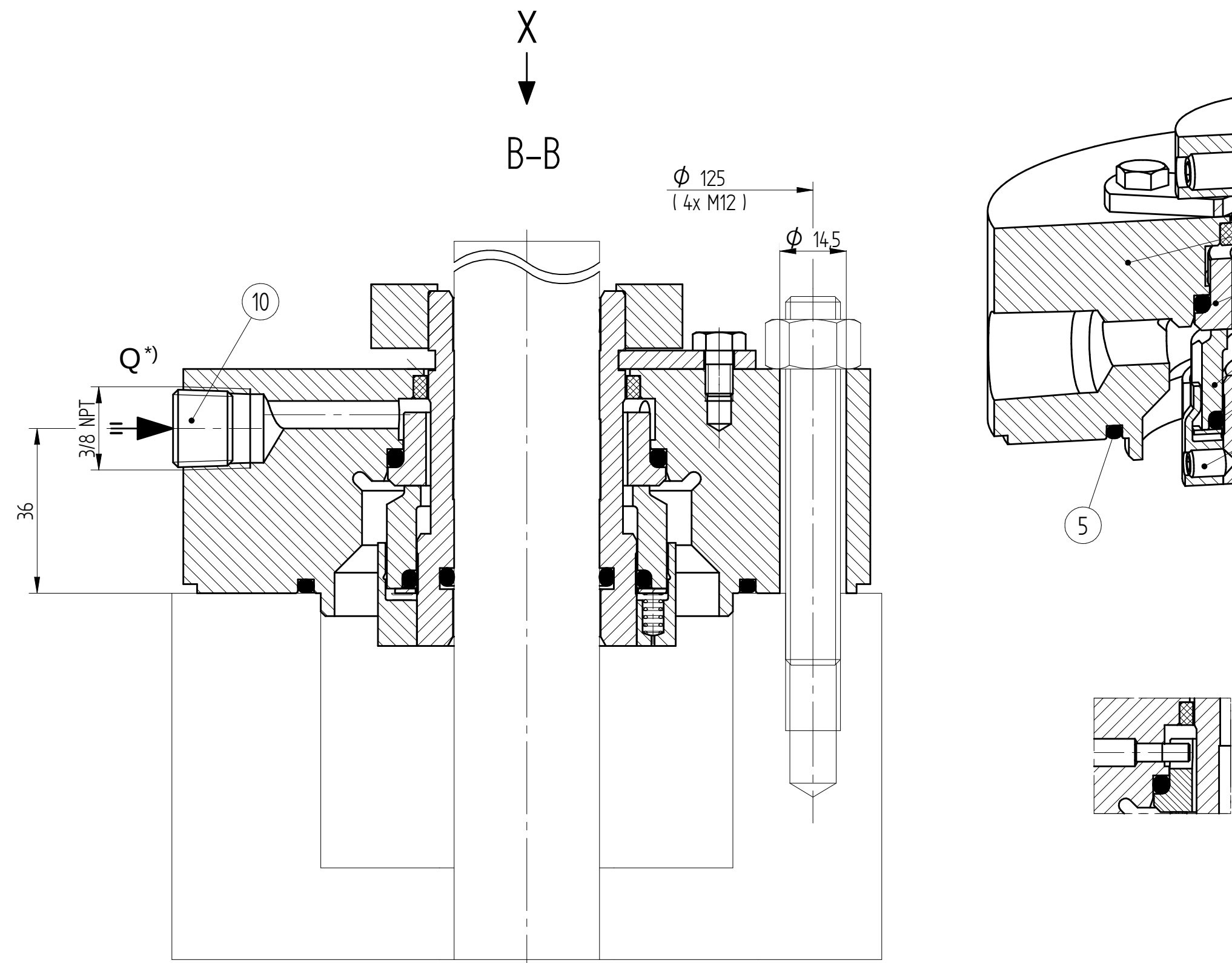
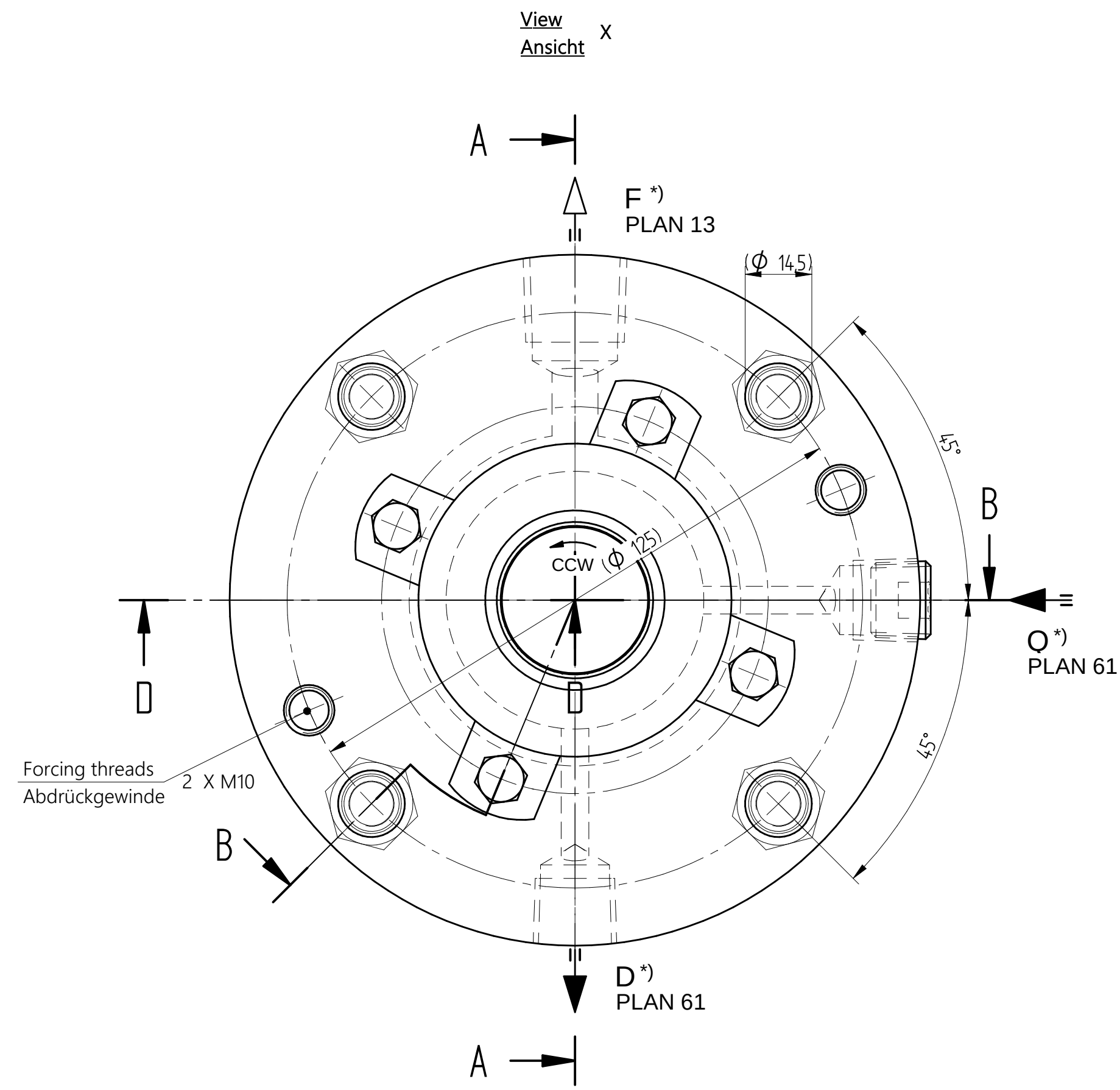
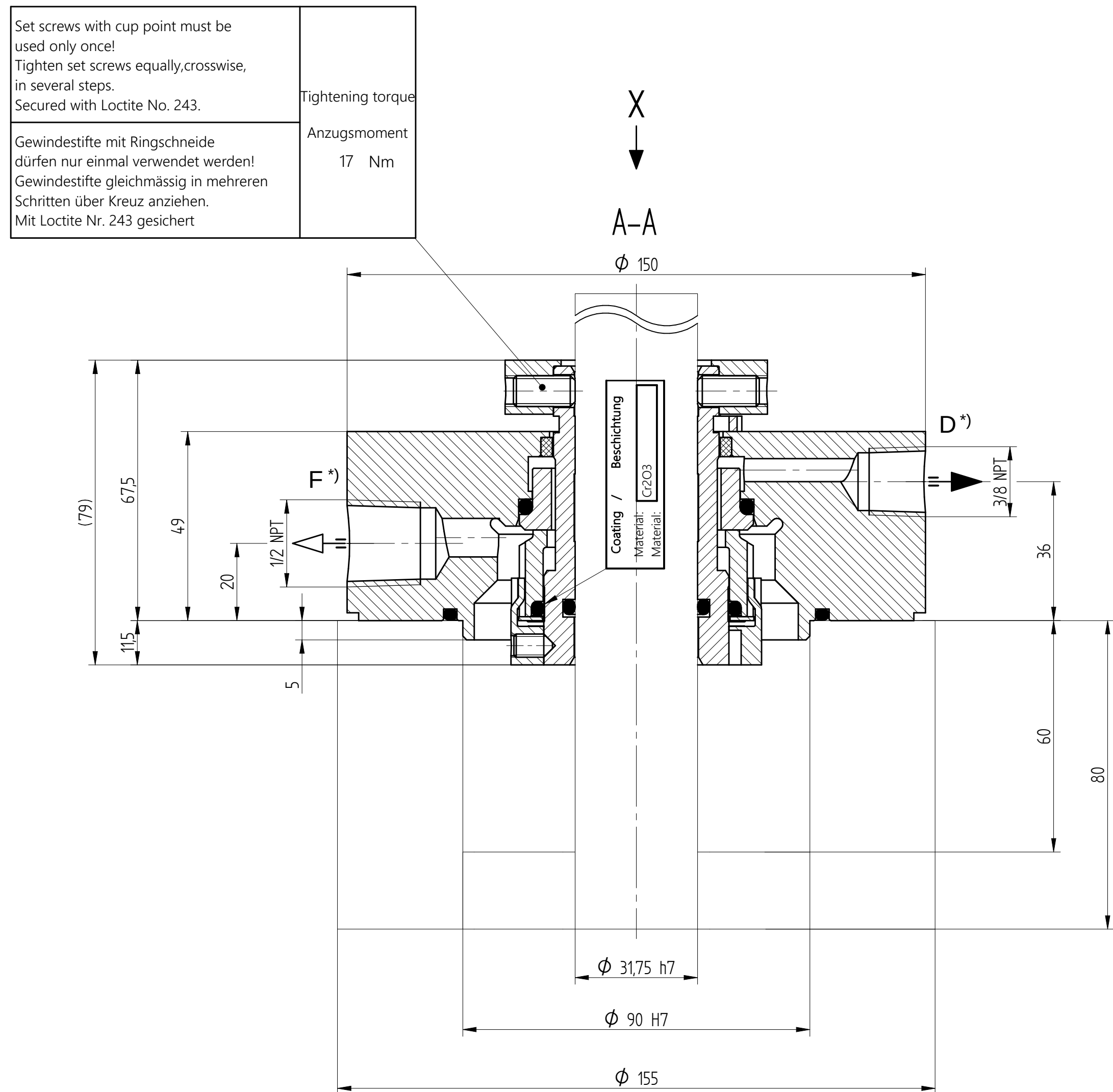


Item	Description
1.1.1	Seal ring
1.1.2	Driver
1.1.3	Thrust ring
1.1.4, 1.3, 3, 6	O-Ring
1.1.5, 8	Spring
1.1.6, 13	Set screw
1.2	Mating ring
2	Seal sleeve
4	Gland plate
5	Flow distributor
7	Throttle ring
9	Disc
10	Setting device
11	Hexagon bolt
12	Set ring

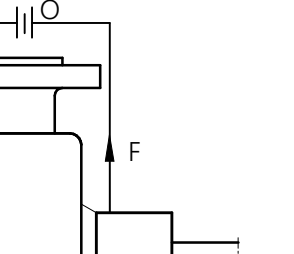
F Flush
Q Quench
D Drain

Dimensions

Dimensions on request.

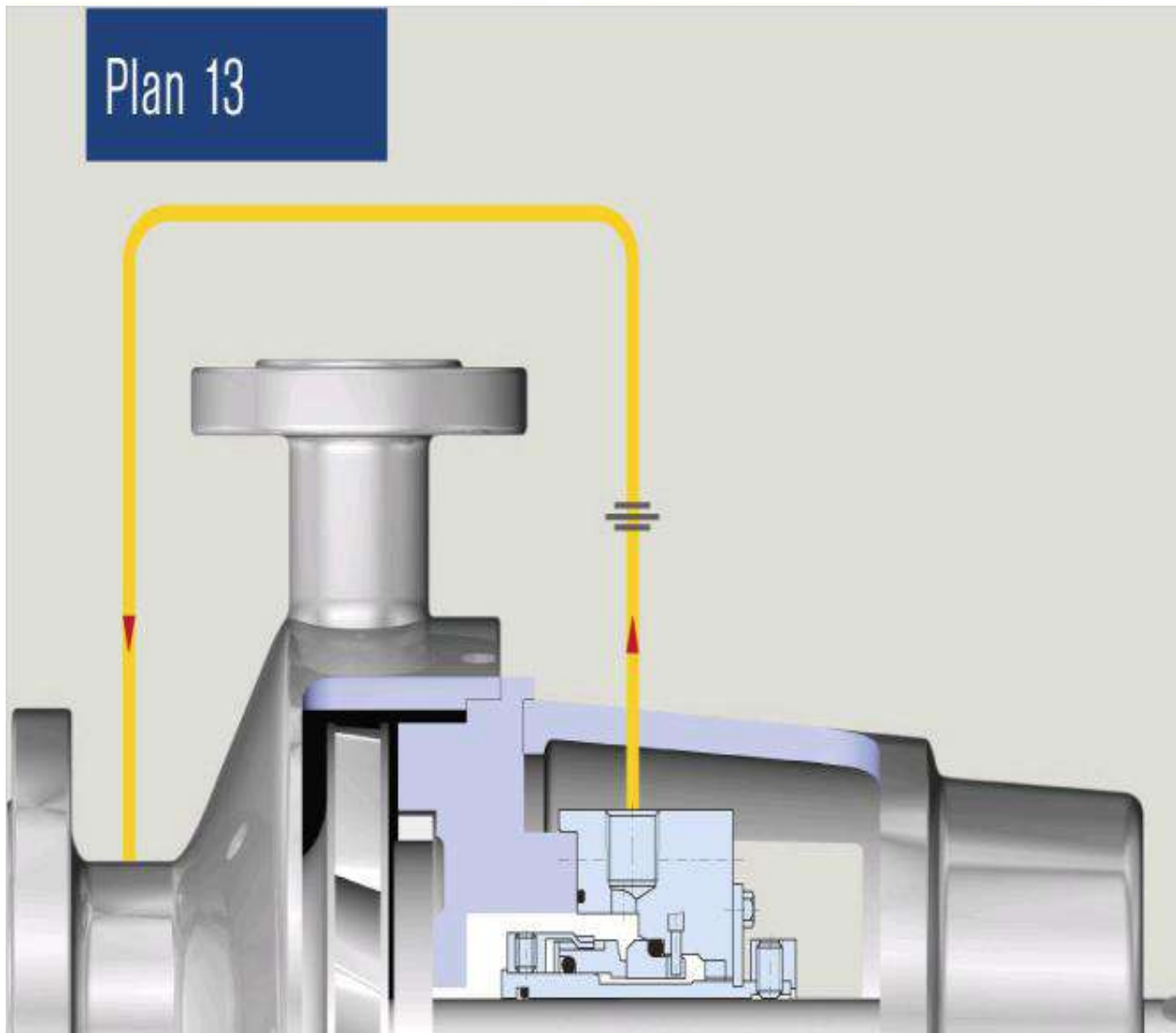


Description Benennung	Description Benennung	Item Pos.	Material Werkstoff	ASTM-Code ASTM-Bezeichnung	Qty. Stueck	Comment Bemerkung
Mechanical seal	Gleitringdichtung	1	Q101KM5G4		1	H75VA4-043-N00
Seal face	Gleitring	1.1	BUKA22-API	Silicon carbide, sintered pressureless	1	
O-Ring	Runddichtring	1.2	K	(ASTM D1418) FFKM	1	
Seat	Gegenring	1.3	BUKA22-API	Silicon carbide, sintered pressureless	1	
O-Ring	Runddichtring	1.4	K	(ASTM D1418) FFKM	1	
Thrust ring	Druckring	1.5	2.4610 (HAST.C4)	ASTM B574 Alloy N06455	1	
Spring	Feder	1.6	2.4819 (HAST.C276)	N10276	12	
Driver	Mitnehmer	1.7	UNSS32760/1.4501	ASTM A276 Type S 32760	1	
Set screw	Gewindestift	1.8	2.4610/2.4819	ASTM B574 Alloy N06455/ASTM B574 Alloy N10276	3	
Shaft sleeve	Wellenhulse	2	UNSS32760/1.4501+CR203	ASTM A276 Type S 32760+ Cr203 Coating	1	
O-Ring	Runddichtring	3	K	(ASTM D1418) FFKM	1	
Cover	Deckel	4	UNSS32760/1.4501	ASTM A276 Type S32760	1	
O-Ring	Runddichtring	5	K	(ASTM D1418) FFKM	1	
Assembly fixture	Montagelehre	6	1.4571	ASTM A276/A479 Type 316Ti	4	
Hexagon bolt	Sechskantschraube	7	2.4610/2.4819	ASTM B574 Alloy N06455/ASTM B574 Alloy N10276	4	
Set ring	Stellring	8	UNSS32760/1.4501	ASTM A276 Type S 32760	1	
Set screw	Gewindestift	9	2.4610/2.4819	ASTM B574 Alloy N06455/ASTM B574 Alloy N10276	6	
Screw plug	Verschlusschraube	10	2.4610/2.4819	ASTM B574 Alloy N06455/ASTM B574 Alloy N10276	1	

API 682 4th edition	Code: 21A-FIQ-032-13/61	Configuration: ICW-FX
	Schlüssel:	Konfiguration:
	Static pressure test acc. API 682 with air at 1,7 bar(g)	
	Druckprobe nach API 682 mit Luft bei (0,17MPa(g); 25 psi(g); 0,17N/mm²(g); 1,7 kg/cm²(g))	
API PLAN 13	Seal type and arrangement complies with API 682	
	Dichtungstyp und Anordnung entspricht der API 682	
	 Remove red plug. Install metal plugs or piping as specified on seal drawing.	
	 Rote Verschlussstopfen entfernen. Metallische Stopfen installieren oder Verrohrung gemäss Vorgabe auf der Dichtungszeichnung.	
API 61	Schematic is for reference only as shown in API 682.	
	Dieses Schaubild gemäß API 682 dient nur als Referenz.	
		Recirculation from the seal chamber through a flow control orifice and back to the pump suction or pump suction piping. Zirkulation vom Dichtungsraum durch Blende zum Saugstutzen bzw. zur Pumpensaugleitung
	Flush fluid flow rate at max. temp. rising Spülmediumdurchfluss bei max. Temp.erhöhung	F Flush O Flow control orifice F Flush O Durchflussregelblende
Remarks	Material code Q1Q1KM5G4	Dynamic sealing pressure rating 580 psi (g)
	Werkstoffschlüssel	Dynamischer Auslegungsdruk
	Allowable temperature min. 30.2 °F	Static sealing pressure rating 870 psi (g)
	Zulässige Temperatur max. 356	Statischer Auslegungsdruk
Operating Data	Medium OILY WATER (TDS 279393)	Pumping temperature min. 95 °F
	Medium	Medium Temperatur max. 122
	Suction pressure min. - psi (a)	Density min. 0.81 specific
	Saugdruck max. 14.7	Dichte max. 1.05
Machine Data	Discharge pressure min. - psi (a)	Viscosity min. 5.6 cP
	Enddruck max. -	Viskosität min. -
	Seal chamber pressure min. 47 psi (a)	Sp.heat fluid min. -
	Druck an der Dichtung max. 147	Wär.Kap. Medium min. -
Seal installation	Max. static sealing pressure (MSSP) - psi (a)	Speed min. 1775 rpm
	Max. stat. Druck an der Dichtung -	Drehzahl min. -
	MAWP - pump 258.3 psi (a)	Speed max. -
	MAWP - Pumpe 258.3	Drehzahl max. -
General tolerances according to ISO 216	Item no. Q04-G-0802 A/B	Manufacturer Sulzer Saudi Pump Company (fo
	Item-Nr.	Maschinenhersteller
	Manufacture model Type / Größe	Stuffing box no. 2-204506534
	End user Saudi Aramco, Saudi Arabia	Einbauraum Nr.
Seal installation	Location Saudi-Arabien	Ambient temp. min. 0 °C
	Ort	Umgebungstemp. max. 0 °C
	-	Direction of shaft rotation seen from drive end
	-	Drehrichtung der Welle vom Antrieb gesehen
Seal installation	max. allowed axial movement of the shaft to the housing	± 1.5 mm (± 0.059 inches)
	max. zul. Axialversatz der Welle zum Gehäuse	
	*) Pipe connections marked correspondingly	
	*) Anschlüsse durch entsprechende Beschriftung gekennzeichnet	
Seal installation	Before startup, the operating manual and the safety notes have to be read and observed!	
	Vor Inbetriebnahme die Betriebsanleitung und die Sicherheitshinweise lesen!	
	General tolerances according to ISO 216	Nominal dimensions (mm) 0.5 - 6 ± 0.1
	Freimastst. nach ISO 2768-m	Tolerances Abmasse 0.5 - 6 ± 0.1
Seal installation	Format / Form A1	Projection Method I
	ISO 216	Method of indicating surface acc. to ISO 1302 (µm)
	No. / Nr.	Tolerances acc. to ISO 2768-mk
	Crit. / Krit.	Refer to protection notice
Seal installation	Modification Date Name	ISO 16016
	Rev.1 17.08.2020 Ganesh K.	ISO 16016
	Drawn Gezeichnet	ISO 16016
	Checked Geprüft	ISO 16016
Seal installation	Scale Masstab 1:1	EagleBurgmann
	Tolerance Toleranz ISO 1015	EagleBurgmann Germany GmbH & Co. KG
	Drawing No. / Zeichnungs-Nr. H75VA4-032-S700-DE003A1	Aussere Sauerländer Strasse 6-10
	Originated from / Entstanden aus H75VA4-030-S700-DE000MA	82375 Wolfarthhausen, Germany

API plan 13

Plan 13



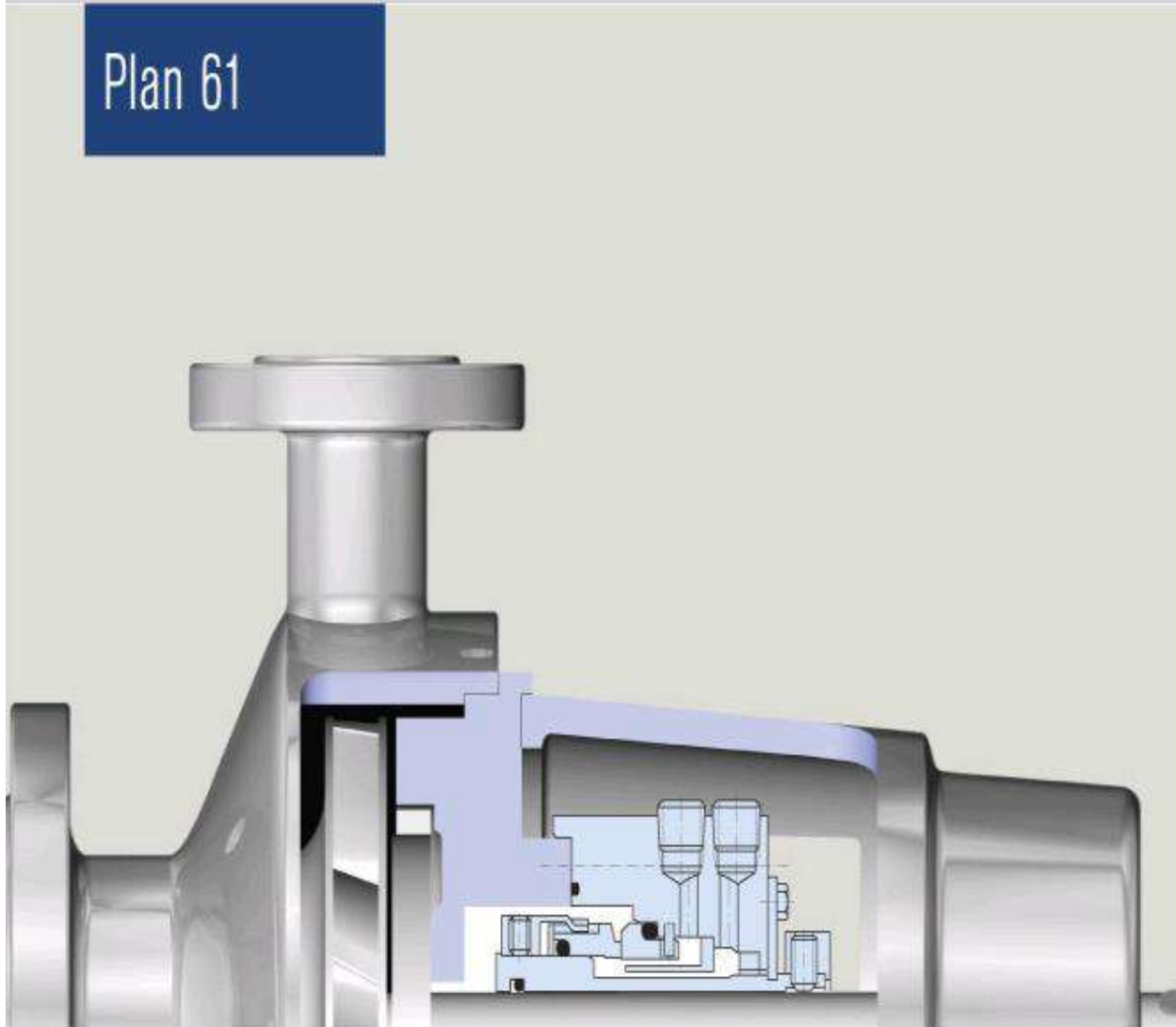
Circulación de la cámara del sello a través de una placa de orificio hacia el lado de succión

Uso:

- para fluidos limpios y uso general
- para bombas verticales
- para bombas con pistones / discos descargados

API plan 61

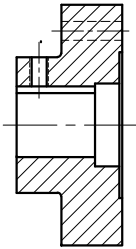
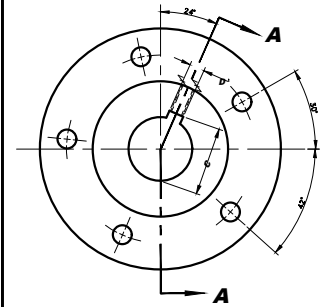
Plan 61



Tapones en las conexiones por el fabricante, no tiene requireimientos especiales.

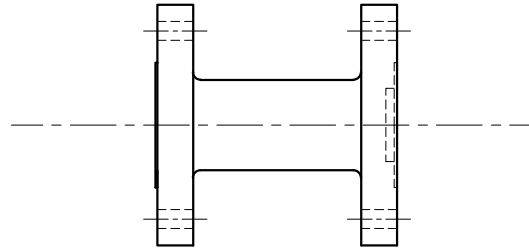
Uso:

- para uso futuro del Plan 62



CORTE A-A

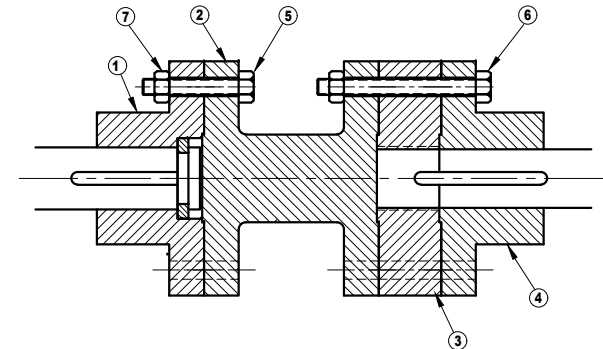
PIEZA No. 1



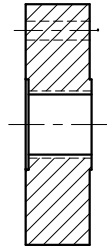
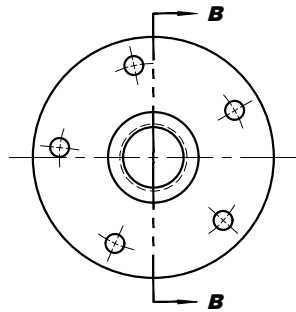
PIEZA No. 2

MAMELON LADO MOTOR								
DIMENSIONES PARA MAQUINADO								
Ø FLECHA	AB	BD	C	D	ED	FB	G	
2.375"								
2.250"								
2.125"								
2.000"								
1.875"								
1.750"								
1.625"								
1.500"								

LISTA DE MATERIALES				
PDA.	CANT.	NO. PZA.	DESCRIPCION	MATERIAL
01	01	1	BARRA MACIZA	ASTM A-434 CL.-BB GR.4140
02	01	2	BARRA MACIZA	ASTM A-434 CL.-BB GR.4140
03	01	3	BARRA MACIZA	ASTM A-434 CL.-BB GR.4140
04	01	4	BARRA MACIZA	ASTM A-434 CL.-BB GR.4140
05	01	S/N	BARRA MACIZA	ASTM A-434 CL.-BB GR.4140
06	05	5	TORNILLO C/HEX.	ACERO 5 (SUPERIOR)
07	05	6	TORNILLO C/HEX.	ACERO 5 (INFERIOR)
08	10	7	TUERCA HEXAGONAL	ACERO 5
09	10	S/N	ROLDANA DE PRESION	ACERO 5
10	02	S/N	PRISIONERO ALIEN	ACERO 5

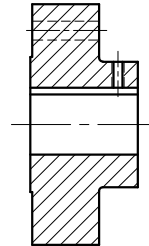
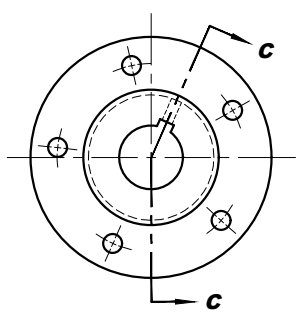


ENSAMBLE GENERAL



CORTE B-B

PIEZA No. 3



PIEZA No. 4

CORTE C-C

1/16" = 125 A MENOS QUE SE ESPECIFIQUE OTRA COSA. DIBUJO REALIZADO EN AUTO CAD

REVISIONES: 0.1	TOLERANCIAS PARA DIMENSIONES NO ESPECIFICADAS				A C A B A D O S		FABRIQUESE EN LA(S) SIG(S) METALURGIAS		 BOMBAS INTERNACIONALES MEXICANAS S.A. de C.V.	REFERENCIA:				
	FRACCIONARIAS ±1/64"				cf	DESVASTE	1000	DESCRIPCION: COPLE RIGIDO		MODELO :	PESO :			
	DECIMALES ±0.005"				rf	ASPERO	500			DIBUJO :				
	ANGULARES 15 MIN				f	REGULAR	125			APROBO :				
					sf	FINO	32			No. DE DIBUJO :				
				ff	PULIDO	16	ACOTACIONES:		ESCALA:	FECHA:	REVISO:	BX9W—		
								PULGADAS.		SIN.	.	.		

NIDEC MOTOR CORPORATION

12443 Olive Boulevard, Suite #100
Creve Coeur, MO 63141



DATE: 6/16/2026

P.O. NO.:

HT42

Order/Line NO.:

27957 MN 100

TO:

Model Number: HT42
Catalog Number: CNT10P2EE
VSS Normal Thrust TEFC Config.
CONF,MOTOR,VSS-NT TEFC

REVISIONS:
(NONE)

**ALL DOCUMENTS HEREIN ARE CONSIDERED CERTIFIED BY NIDEC MOTOR CORPORATION.
THANK YOU FOR YOUR ORDER AND THE OPPORTUNITY TO SERVE YOU.**

Features:

Horsepower 00010.00~00000.00 ~ KW: 7.46
Enclosure TEFC
Poles 04~00 ~ RPM: 1800~0
Frame Size 215~HP
Phase/Frequency/Voltage.. 3~060~230/460 (Usable at 208V)
Winding Type Random Wound
Service Factor 1.15
Insulation Class Class "F" ~ Insulife 2000
Altitude In Feet (Max) .. 3300 Ft.(1000 M)
Ambient In Degree C (Max) +40 C
Efficiency Class Premium Efficiency
Application Centrifugal Pump
Customer Part Number
Base Diameter (Inches) 10
Pricebook Thrust Value (lbs).. 640
Customer Down Thrust (lbs) ... 640
Customer Shutoff Thrust (lbs).
Up Thrust (lbs): ~
Secondary Rating:
10 Horsepower ~ 50 Hertz ~ 200/400 Volts
"AK" Dimension (Inches).. NA
Shaft Dimensions:~U=1.125 ~ AH/V=2.750
KEYWAY=0.250 ~ ES=1.280
Temperature Rise (Sine Wave): "B" Rise @ 1.0 SF (Resist)
Design Letter B
Starting Method Direct-On-Line Start
Duty Cycle Continuous Duty
Efficiency Value 91.7 % ~ Typical
Load Inertia: NEMA ~ Standard Inertia: 51 LB-FT²
Number Of Starts Per Hour: NEMA
Motor Type Code TVCS
Rotor Inertia (LB-FT²)967 LB-FT²
Qty. of Bearings PE (Shaft) 1
Qty. of Bearings SE (OPP) 1
Bearing Number PE (Shaft) 6309-J/C3
Bearing Number SE (OPP) 6208-J/C3

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NIDEC MOTOR CORPORATION

12443 Olive Boulevard, Suite #100
Creve Coeur, MO 63141



DATE: 6/16/2026

P.O. NO.: HT42
Order/Line NO.: 27957 MN 100

TO:

Model Number: HT42
Catalog Number: CNT10P2EE
VSS Normal Thrust TEFC Config.
CONF,MOTOR,VSS-NT TEFC

REVISIONS:
(NONE)

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THANK YOU FOR YOUR ORDER AND THE OPPORTUNITY TO SERVE YOU.**

Accessories:

Shaft Slinger - Both Ends
Brass Drain - Lower Bracket
Corro-Duty
Ground Lug In Conduit Box
Standard Leadtime: NA
F.O.B.:
PRIMARY GROUND
LOCATED IN MAIN
CONDUIT BOX
~
~

USE THE DATA PROVIDED BELOW TO SELECT THE APPROPRIATE DIMENSION PRINT

Horsepower	10
Pole(s)	04
Voltage(s)	460-230 / 400-200
Frame Size	215HP
Shaft U Diameter	1.125
Outlet Box AF	2.63
Outlet Box AA	1.00

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EFFECTIVE:
01-MAY-24

SUPERSEDES:
07-SEP-23

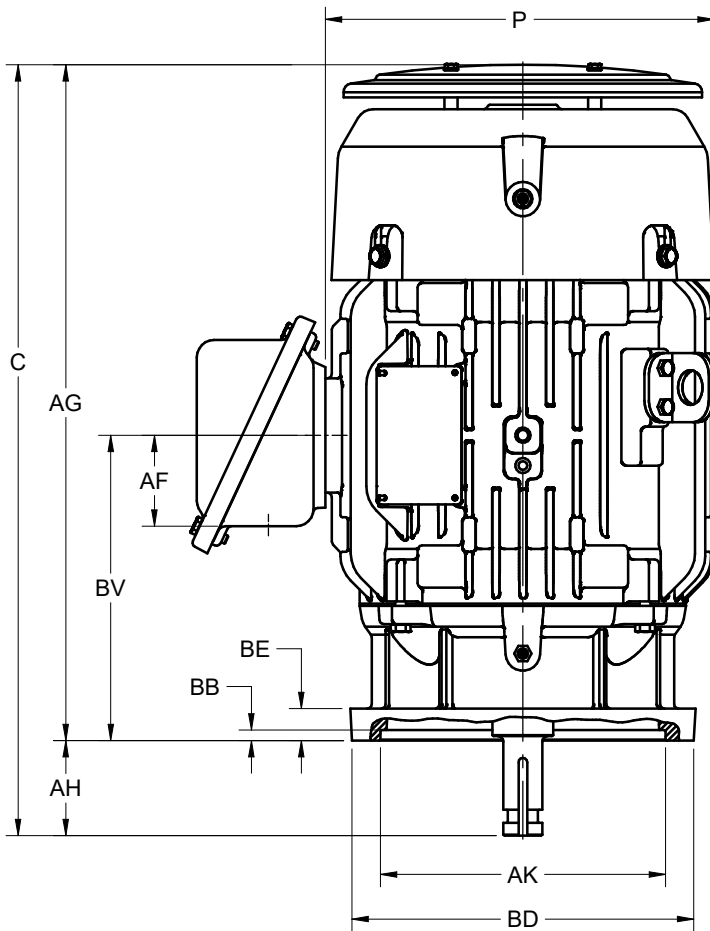
DIMENSION PRINT

WORLD MOTOR VERTICAL CORRODUTY

FRAME: 210HP
BASIC TYPE: TVC

PRINT:
10-1006

SHEET:
1 OF 1



ALL DIMENSIONS ARE IN INCHES AND MILLIMETERS

UNITS	C	P ²	U -.0005	V MIN	AA
IN	22.31	11.50	1.1250	2.75	1 NPT
MM	567	292	28.575	70	

UNITS	AB	AC	AF	AG	AH
IN	9.56	7.38	2.63	19.56	2.75
MM	243	187	67	497	70

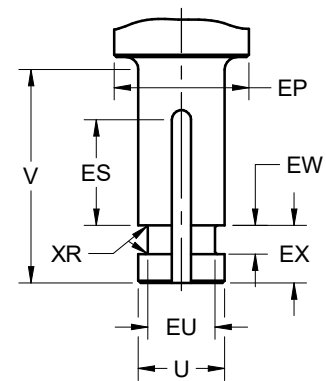
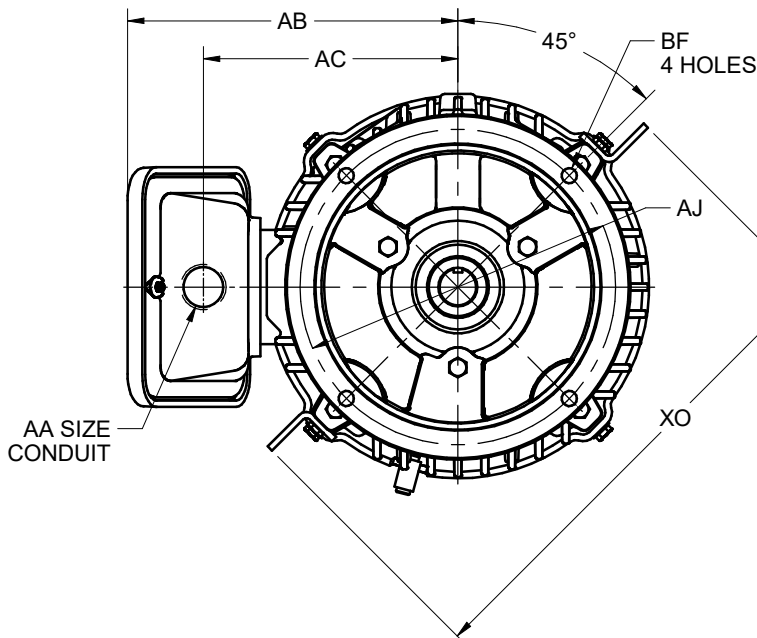
UNITS	AJ	AK +.003	BB MIN	BD MAX	BE
IN	9.125	8.250	.19	10.00	.94
MM	231.78	209.55	5	254	24

UNITS	BF	BV	EP MIN	ES MIN	EU -.005
IN	.44	8.84	1.375	1.28	.875
MM	11	225	34.93	33	22.23

UNITS	EW +.002	EX -.005	XO	XR	SQ KEY
IN	.375	.750	14.31	.03	.250
MM	9.53	19.05	363	1	6.35

TOLERANCES

FACE RUNOUT	.004 TIR
PERMISSIBLE ECCENTRICITY OF MOUNTING RABBET	.004 TIR
PERMISSIBLE SHAFT RUNOUT	.002 TIR



1. DIMENSIONS MAY VARY .25" DUE TO CASTING AND/OR FABRICATION VARIATIONS
2. LARGEST MOTOR WIDTH
3. STANDARD AS SHOWN WITH CONDUIT OPENING DOWN. CONDUIT OPENING MAY BE LOCATED IN STEPS OF 90°
4. TOLERANCES ARE SHOWN IN INCHES ONLY

10-1006/B

Nidec Motor Corporation
St. Louis, Missouri

INFORMATION DISCLOSED ON THIS DOCUMENT
IS CONSIDERED PROPRIETARY AND SHALL NOT BE
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CONSENT OF NIDEC MOTOR CORPORATION



ISSUED BY
R. TIMMERMANN
APPROVED BY
E. TREVINO

IHP_DP_NMCA (MAR-2011) SOLIDEDGE

NAMEPLATE DATA

CATALOG NUMBER:		CNT10P2EE		NAMEPLATE PART #:		2106033-007	
MODEL	HT42	FR	215HP	TYPE	TVCS	ENCL	TEFC
SHAFT END BRG		6309-J/C3 - QTY 1		OPP END BRG		6208-J/C3 - QTY 1	
PH	3	MAX AMB	40 C	ID#			
INSUL CLASS	F	Asm. Pos.			DUTY	CONT	
HP	10	RPM	1770	HP	10	RPM	1465
VOLTS	460	230		VOLTS	400	200	
FL AMPS	12.8	25.7		FL AMPS	14.9	29.7	
SF AMPS	14.5	29.1		SF AMPS			
SF	1.15	DESIGN	B	CODE	H	SF	1.00
NEMA NOM EFFICIENCY	91.7	NOM PF	79.5	KiloWatt	7.46	NEMA NOM EFFICIENCY	90.2
GUARANTEED EFFICIENCY	90.2	MAX KVAR	3.9	HZ	60	NOM PF	80.0
						GUARANTEED EFFICIENCY	88.5
						MAX KVAR	3.8
						HZ	50

HAZARDOUS LOCATION DATA (IF APPLICABLE):

DIVISION		CLASS I		GROUP I	
TEMP CODE		CLASS II		GROUP II	



VFD DATA (IF APPLICABLE):

VOLTS		AMPS	
TORQUE 1		TORQUE 2	
VFD LOAD TYPE 1		VFD LOAD TYPE 2	
VFD HERTZ RANGE 1		VFD HERTZ RANGE 2	
VFD SPEED RANGE 1		VFD SPEED RANGE 2	
SERVICE FACTOR		FL SLIP	
NO. POLES	4	MAGNETIZING AMPS	5.7
VECTOR MAX RPM		Encoder PPR	
Radians / Seconds		Encoder Volts	

TEAO DATA (IF APPLICABLE):

HP (AIR OVER)		HP (AIR OVER M/S)		RPM (AIR OVER)		RPM (AIR OVER M/S)	
FPM AIR VELOCITY		FPM AIR VELOCITY M/S		FPM AIR VELOCITY SEC			

ADDITIONAL NAMEPLATE DATA:

Decal / Plate	WD=109144	Customer PN	
Notes		Non Rev Ratchet	
Max Temp Rise	80C RISE/RES@FL 60HZ	OPP/Upper Oil Cap	GREASE
Thermal (WDG)		SHAFT/Lower Oil Cap	GREASE
Altitude		Usable At	
Regulatory Notes		Regulatory Compliance	CC 030A
COS		Marine Duty	
Balance		Arctic Duty	
3/4 Load Eff.	92.6	Inrush Limit	
Motor Weight (LBS)	210	Direction of Rotation	
Sound Level		Special Note 1	
Vertical Thrust (LBS)	640	Special Note 2	
Thrust Percentage		Special Note 3	
Bearing Life		Special Note 4	
Starting Method		Special Note 5	
Number of Starts		Special Note 6	
200/208V 60Hz Max Amps		SH Max. Temp.	
190V 50 hz Max Amps		SH Voltage	
380V 50 Hz Max Amps		SH Watts	
NEMA Inertia		Load Inertia	
Sumpheater Voltage		Sumpheater Wattage	
Special Accessory Note 1		Special Accessory Note 16	
Special Accessory Note 2		Special Accessory Note 17	
Special Accessory Note 3		Special Accessory Note 18	
Special Accessory Note 4		Special Accessory Note 19	
Special Accessory Note 5		Special Accessory Note 20	
Special Accessory Note 6		Special Accessory Note 21	
Special Accessory Note 7		Special Accessory Note 22	
Special Accessory Note 8		Special Accessory Note 23	
Special Accessory Note 9		Special Accessory Note 24	
Special Accessory Note 10		Special Accessory Note 25	
Special Accessory Note 11		Special Accessory Note 26	
Special Accessory Note 12		Special Accessory Note 27	
Special Accessory Note 13		Special Accessory Note 28	
Special Accessory Note 14		Special Accessory Note 29	
Special Accessory Note 15		Special Accessory Note 30	
Heater in C/B Voltage		Heater in C/B Watts	
Zone 2 Group		Division 2 Service Factor	
Note 1	CL.I GR.A B C D	Note 2	CL.II GR.F
Note 3	T3 TEMP CODE DIV 2	Note 4	AT 1.0 S.F.
Note 5	USABLE AT:	Note 6	
Note 7		Note 8	208V 0 AMPS
Note 9		Note 10	
Note 11		Note 12	
Note 13		Note 14	
Note 15		Note 16	
Note 17		Note 18	
Note 19		Note 20	
Note 21		Note 22	

NIDEC MOTOR CORPORATION
ST. LOUIS, MO

TYPICAL NAMEPLATE DATA
ACTUAL MOTOR NAMEPLATE LAYOUT MAY VARY
SOME FIELDS MAY BE OMITTED

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MOTOR PERFORMANCE

MODEL NO.	CATALOG NO.	PHASE	TYPE	FRAME	
HT42	CNT10P2EE	3	TVCS	215HP	
ORDER NO.		27957		LINE NO.	
MPI:		305771	305772	329569	305774
HP:		10	10	10	10
POLES:		4	4	4	4
VOLTS:		460	230	208	400
HZ:		60	60	60	50
SERVICE FACTOR:		1.15	1.15	1	1
EFFICIENCY (%):					
S.F.		91.5	91.5		
FULL		91.7	91.7	90.2	90.2
3/4		92.6	92.6	91.8	91.9
1/2		92	92	91.8	91.9
1/4		87.8	87.8	88.8	88.5
POWER FACTOR (%):					
S.F.		81	81		
FULL		79.5	79.5	82.7	80
3/4		74.5	74.5	80.4	75.2
1/2		64	64	73.2	64.6
1/4		42.6	42.6	53.8	42.8
NO LOAD		4.8	4.8	5.6	4.3
LOCKED ROTOR		39.1	39.1	37.8	41.8
AMPS:					
S.F.		14.5	29.1		
FULL		12.8	25.7	27.5	14.9
3/4		10.2	20.3	20.9	11.7
1/2		8	15.9	15.3	9.1
1/4		6.3	12.5	10.8	7.1
NO LOAD		5.7	11.4	8.9	6.5
LOCKED ROTOR		88	176	151.8	90.3
NEMA CODE LETTER		H	H	F	H
NEMA DESIGN LETTER		B	B	#	#
FULL LOAD RPM		1770	1770	1760	1465
NEMA NOMINAL / EFFICIENCY (%)		91.7	91.7	90.2	90.2
GUARANTEED EFFICIENCY (%)		90.2	90.2	88.5	88.5
MAX KVAR		3.9	3.9	2.7	3.8
AMBIENT (°C)		40	40	40	40
ALTITUDE (FASL)		3300	3300	3300	3300
SAFE STALL TIME-HOT (SEC)		24	24	30	23
SOUND PRESSURE (DBA @ 1M)		65	65	70	61
TORQUES:					
BREAKDOWN{% F.L.}		268	268	226	240
LOCKED ROTOR{% F.L.}		219	219	175	193
FULL LOAD{LB-FT}		29.7	29.7	29.8	35.9

NEMA Nominal and Guaranteed Efficiencies are up to 3,300 feet above sea level and 25 ° C ambient

The Above Data Is Typical, Sinewave Power Unless Noted Otherwise

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ST. LOUIS, MO

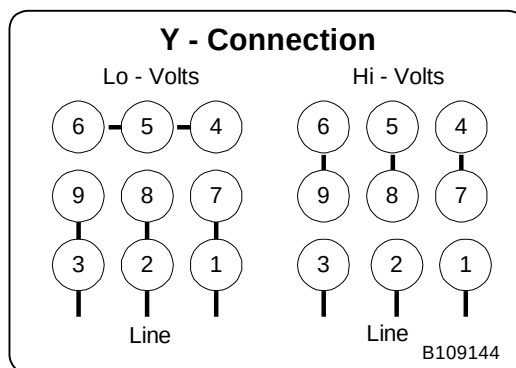
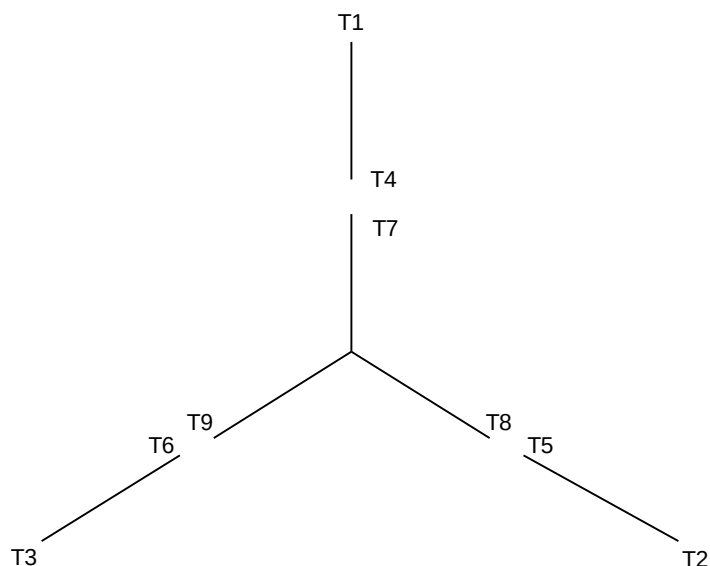


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B109144

Motor Wiring Diagram
9 Lead, Dual Voltage (WYE Conn.)

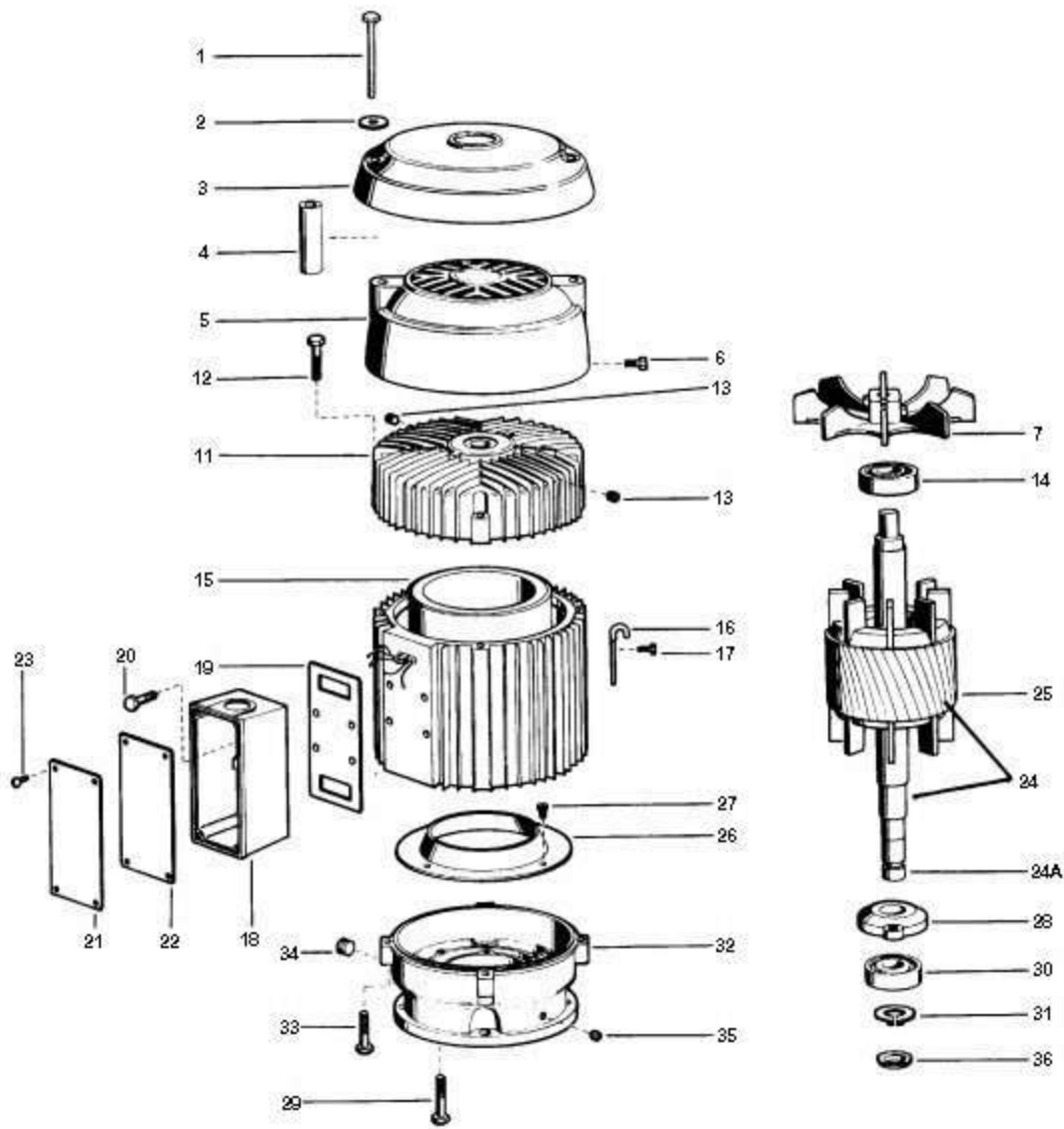


To reverse direction of rotation interchange connections L1 and L2.

Each lead may have one or more cables comprising that lead.
In such case each cable will be marked with the appropriate lead number.

RENEWAL PARTS

FRAMES 182 THRU 365 - OPEN DRIPPROOF, TOTALLY ENCLOSED AND EXPLOSION PROOF
 TYPES: AV, AV9, AVE, AVI, LV, LV4, LV9, LVC4, LVC9, LVCS4, LVCS9, LVE, LVE4, LVE9, LVI4, LVS4, LVS9, NVC4, NVC9,
 NVCE, NVCS, NVE, TV, TV4, TV9, TVC, TVC4, TVC9, TVCE, TVCE9, TVCI, TVCI4, TVCI9, TVCS, TVCS4, TVCS9, TVE,
 TVE4, TVE9, TVI, TVI4, TVI9, TVS, TVS4, TVS9
 SOLIDSHAFT MOTORS



ITEM NO.	QTY	NAME OF PART
1	3	Screw (Used on 213 thru 256, type AV & TV)
2	3	Washer (Canopy Cap) (Used on 254 thru 256, type AV & TV)
3	1	Canopy Cap (Not used on frames 182 - 184 types AV, TV, & LV, frames 213 - 215 type LV, frames 284-286 types TV & LV)
4	3	Spacer (Used on frames 213 thru 256, type AV & TV)
5	1	Fan Cover
6	4	Screw (Qty. 8 Hex Nut on frame 280 & up)

WARNING:

Any disassembly or repair work on explosionproof motors will void the Underwriters Laboratories, Inc. label unless done by the manufacturer, or a facility approved by the Underwriters Laboratories, Inc. Refer to your nearest sales office for assistance.

BEARINGS:

Refer to motor nameplate for the bearing numbers.

PRICES:

Parts stocking distributors: refer to renewal parts numerical index. All Others: refer to your nearest parts distributor.

RENEWAL PARTS

FRAMES 182 THRU 365 - OPEN DRIPPROOF, TOTALLY ENCLOSED AND EXPLOSION PROOF
 TYPES: AV, AV9, AVE, AVI, LV, LV4, LV9, LVC4, LVC9, LVCS4, LVCS9, LVE, LVE4, LVE9, LVI4, LVS4, LVS9, NVC4, NVC9,
 NVCE, NVCS, NVE, TV, TV4, TV9, TVC, TVC4, TVC9, TVCE, TVCE9, TVCI, TVCI4, TVCI9, TVCS, TVCS4, TVCS9, TVE,
 TVE4, TVE9, TVI, TVI4, TVI9, TVS, TVS4, TVS9
 SOLIDSHAFT MOTORS

ITEM NO.	QTY	NAME OF PART
7	1	Fan
8	2	Screw (Fan) (Not Illustrated)
9	2	Washer (Fan) (Not Illustrated) (Used on frames 182 thru 256, types AV, TV & LV)
10	2	Square Nut (Fan) (Not Illustrated) (Used on frames 182 thru 256, types AV, TV & LV)
11	1	Bracket (Upper)
12	4	Screw (Used on frames 182-184, types AV, TV, & LV, frames 213-256 types AV & TV)
13	1	Pipe Plug (Not used on frames 324 thru 326)
14	1	Bearing (Upper) (Refer to section 775)
	1	Bearing Cap (Upper) (Not Illustrated) (Not used on frames 182 thru 256, types AV & TV)
	3	Screw (Not used on frames 182 thru 256, types AV & TV)
15	1	Wound Stator Assembly
16	2	Lifting Hook (Eyebolt on some units)
17	4	Screw (Lifting Hook) (Used on frames 182-184, types AV, TV, & LV, frames 213-256 types AV & TV)
18	1	Outlet Box Base
19	4	Gasket (Used on frames 180 thru 250, type AV & TV, frames 280 thru 360 type TV)
20	1	Screw (Outlet Box Base)
21	1	Outlet Box Cover
22	1	Gasket (Not used on type LV)
23	4	Screen (Outlet Box Base)
24	1	Rotor Assembly (Includes items 24A & 25)
24A	1	Shaft
25	1	Rotor Core
26	1	Air Deflector (Not used on frames 180 thru 250, type LV, frames 280 thru 320 type AV)

WARNING:

Any disassembly or repair work on explosionproof motors will void the Underwriters Laboratories, Inc. label unless done by the manufacturer, or a facility approved by the Underwriters Laboratories, Inc. Refer to your nearest sales office for assistance.

BEARINGS:

Refer to motor nameplate for the bearing numbers.

PRICES:

Parts stocking distributors: refer to renewal parts numerical index. All Others: refer to your nearest parts distributor.

ITEM NO.	QTY	NAME OF PART
27	1	Screw (Air Deflector) (Not used on frames 180 thru 250, type LV, frames 280 thru 320 type AV)
28	1	Bearing Cap (Lower) (Qty. 2 on frame 180 type LV)
29	2	Screw (Bearing Cap (Qty. 3 on TV & LV)
30	1	Bearing (Lower) (Refer to section 775)
31	1	Snap Ring (Not used on frames 180 thru 210, type LV)
32	1	Unimount Bracket
33	4	Screw (Qty. 8 on frame 280 and up)
34	1	Pipe Plug (Fill) (Not used on frame 280 thru 360, type TV & LV)
35	1	Pipe Plug (Drain)
36	1	Water Deflector (Not used on frame 180, type LV)
	1	Pipe Nipple (Fill) (Not Illustrated) (Not used on frame 180, type AV, TV & LV)
	1	Pipe Coupling (Fill) (Not Illustrated) (Not used on frame 180, type AV, TV & LV)
	1	Grease Fitting (Qty. 2 on frame 280 thru 360) (Not used on frame 180)
	1	Fan Grill (Used on frames 210 thru 250, type LV, frame 280, type TV & LV)
	4	Stud (Qty. 8 on frame 320 & up) (Not Illustrated) (Not used on frame 180; frame 210 & 250 types AV & TV)
	8	Hex Nut (Not Illustrated) (Not used on frame 180; frame 210 & 250 types AV & TV)
	1	Pipe Nipple, drain (Not Illustrated) (Not used on frames 180 thru 210, type AV, TV & LV)
	1	Pipe Cap, Drain (Not Illustrated) (Not used on frames 180 thru 210, type AV, TV & LV)
	1	Woodruff Key (Fan) (Not Illustrated) (Used on frames 320 thru 360, types TV & LV)

reference: Renewal Parts Section 700, Pages 155 & 156

TYPICAL REED CRITICAL FREQUENCY DATA

Note: Motor RCF Test Data can be provided at time of motor shipment through special test.
Please contact your Nidec Motor Corporation representative for more information.

MODEL NO: HT42
CATALOG NO: CNT10P2EE

Frame: 215HP Type: TVCS

REED CRITICAL FREQUENCY:	212	HZ
CENTER OF GRAVITY:	9.25	IN
DEFLECTION @ CENTER OF GRAVITY:	0.00022	IN
UNIT WEIGHT:	188	LBS
BASE DIAMETER:	10	IN
TOLERANCE ON RCF VALUE:	20%	
DATE:	6/16/2026	



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General Information for Integral Horsepower (IHP) Motors on Variable Frequency Drives (VFDs)

Variable Frequency Drives (VFD)

A VFD is a type of controller used to vary the speed of an electric motor. The VFD takes a fixed AC voltage and frequency and allows it to be adjusted in order to get different speeds from the motor. Motor speed can be varied by changing the frequency of the input power waveform. The equation below shows how the frequency affects the speed of a three phase induction motor.

$$\text{Speed} = \frac{120 * \text{Fundamental Input Frequency}}{\text{Number of Motor Poles}}$$

How does a VFD work?

A VFD takes the fixed frequency and voltage sine wave from the power grid or power station and puts it through a few steps in order to allow the VFD user to vary the frequency and in turn control the motor speed. First it rectifies the AC power into DC Power. Because of this step, a term commonly used instead of VFD is inverter. This only describes one step of what the VFD does to the power waveform. Once rectified into a DC voltage the drive sends the power through a set of transistors or switches. These switches can take the DC waveform and by opening and closing at certain speeds and durations can create an output waveform that mimics the sine wave that is required to drive a three phase electric motor. The output wave form is known as a Pulse Width Modulation (PWM) waveform because the waveform is created by multiple pulses of the switches at short intervals.

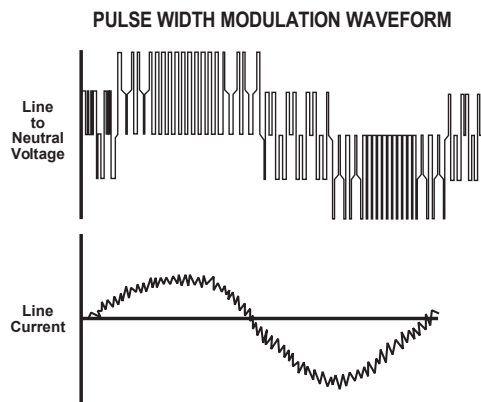


Figure 1 PWM Waveform

What variables should be considered when deciding whether to power a motor with a VFD?

VFD compatibility with motors is complex. As a result, many variables must be considered when determining the suitability of a particular motor for use with a VFD. These variables include:

- Torque requirements (Constant or Variable)
- Speed Range
- Line / System Voltage
- Cable length between the VFD and the motor
- Drive switching (carrier) frequency
- Motor construction

- VFD dv/dt - winding end turn differential in voltage versus differential in time
- High temperatures or high humidity
- Grounding system

Wider speed ranges, higher voltages, higher switching frequencies, insufficient grounding and increased cable lengths all add to the severity of the application and, therefore, the potential for premature motor failure.

How does a VFD affect the motor?

There are many things to consider when a motor is powered using a VFD or PWM power. When a motor is powered by a PWM waveform the motor windings very often see a large differential voltage, either from phase to phase or turn to turn. When the voltage differential becomes large enough it creates a reaction at the molecular level that converts available oxygen into O₃. This phenomenon is called partial discharge or corona. This reaction creates energy in the form of light and heat. This energy has a corrosive effect on the varnish used to protect the motor windings. PWM waveforms can also magnify shaft voltages which lead to arcing across the bearing and causing premature bearing failure. Corrective action must be taken to mitigate these issues that arise when using an electric motor with a VFD.

How do I protect the motor?

Nidec Motor Corporation (NMC) has developed specific motor designs to decrease the harmful affects that a VFD can have on a motor. NMC's INVERTER GRADE® insulation system is the first line of defense against corona and phase to phase faults that can be common when a motor is powered using a PWM waveform. The INVERTER GRADE® insulation system is standard on all of NMC's Inverter Duty products. Along with the INVERTER GRADE® insulation, thermostats are installed as a minimum protection against over heating the motor. Special consideration must also be given to bearings in motors powered by VFD's. In order to create a low resistance path to ground for built up shaft voltages a shaft grounding device can be used. On larger horsepower motors an insulated bearing system should be used in conjunction with the shaft grounding device when installed, to force the stray shaft voltages to ground. The bearing failures are more prominent on motors with thrust handling bearings. NMC has created an Inverter Duty vertical motor line that not only uses the INVERTER GRADE® insulation system, but that also comes standard with a shaft grounding device. On motors that are 100 HP and greater the thrust bearing is also insulated for additional protection.

What does "Inverter Duty" mean?

An Inverter Duty motor should describe a motor that helps mitigate potential failure modes of a motor that is powered by a VFD. Inverter duty motor windings should be able to withstand the voltage spikes per NEMA MG1 Part 31.4.4.2 and protect against overheating when the motor is run at slow speeds. On thrust handling bearings it is apparent that the bearings require additional protection. Inverter Duty vertical motors should have a shaft grounding device to protect the motor bearings from fluting due to voltage discharge through the bearing. On larger motors (100HP and larger) the shaft should also be electrically isolated from the frame in order to aid the shaft grounding ring in discharging the shaft voltages to ground.

*This information applies only to Integral Horsepower (IHP) motors as defined on the Agency Approval page, under UL® & CSA® listings where indicated.

Motor / Inverter Compatibility

Thermal Overloads and Single Phase Motors

Motors with thermal overloads installed may not operate properly on a VFD. The current carrying thermal overload is designed for sine wave power. Operation on a VFD may cause nuisance tripping or potentially not protect the motor as would be expected on line power. Thermostats or thermistors installed in the motor and connected properly to the VFD may provide suitable thermal overload protection when operating on a VFD. (consult codes for installation requirements)

Single phase motors and other fractional horsepower ratings are not designed to be operated on a VFD. Within Nidec Motor Corporation standard products, all motors NEMA[®] 48 frame (5.5" diameter) and smaller are not suitable for VFD applications. Three phase 56 and 143/145 frame applications should be noted on the catalog price page; or if in doubt ask a Nidec Motor Corporation technical representative for recommendations on compatibility with a VFD.

Slow Speed Motors

Motors with a base design of slower than six poles require special consideration regarding VFD sizing and minimizing harmonic distortion created at the motor terminals due to cable installation characteristics. Additional external PWM waveform filters and shielded motor cables designed for PWM power may be required to provide acceptable motor life. Harmonic distortion on the output waveform should be kept to a minimum level (less than 10%) mismatch impedance.

690V Applications

Motors that are rated for 690VAC and that will be powered by 690VAC PWM VFDs require the use of an external filter to limit peak voltage spikes and the use of an INVERTER GRADE[®] motor. Where available, an alternative to using an output filter is to upgrade to a 2300V insulation system.

Low Voltage TITAN[®] Motors

When using 449 frame and larger motors on PWM type VFDs consider the use of an external filter and shielded motor cables designed for PWM power to minimize harmonic distortion and peak voltages at the motor terminals. Harmonic distortion on the output waveform should be kept to a minimum level (less than 10%).

Bearing Currents Related to PWM Waveforms

Protection of the motor bearings from shaft currents caused by common mode voltages is becoming a standard feature on Inverter Duty motor products. Some installations may be prone to a voltage discharge condition through the motor bearings called Electrical Discharge Machining (EDM) or fluting. Vertical HOLLOSHAFT and HOSTILE DUTY World Motor come with grounding devices installed as standard. EDM damage is related to characteristics of the PWM waveform, and the VFD programming, and installations factors.

Bearing Protection on Inverter Duty Vertical Motors

All U.S. MOTORS[®] brand "Inverter Duty" vertical products have a shaft grounding system that allows damaging shaft currents a low resistance path to ground. **Bearings on vertical motors fed by VFD power without this bearing protection are not covered under any warranty.** All other bearing failure is covered per NMC's standard warranty. An electric motor repair shop approved to service U.S. MOTORS[®] brand motors must verify that the cause of the bearing failure was not due to EDM damage.

Guideline For Insulated Anti-Friction Bearings

Bearing insulation is required to prevent circulating shaft currents which can damage bearings. Circulating shaft current can be caused by use of improper power and/or ground cables, improper grounding systems and higher switching frequencies. Finding and correcting the external condition(s) is the responsibility of the system designer or specifying engineer. To prevent circulating shaft current in motors with anti-friction bearings, Nidec Motor Corporation's standard practice is to insulate the non-drive end bearing.

Adjustable Speed Drives produce a common mode voltage condition. To interrupt common mode voltage on induction motors of all sizes, NEMA MG1-2018 Part 31 recommends insulating both bearings. In cases where both anti-friction bearings are insulated, the system designer or specifying engineer should determine whether to apply one or more of the following options to prevent or reduce shaft currents: sinewave filters, line reactors or mechanical devices, such as shaft grounding or an insulated half coupling. Motors with shaft grounding devices are not suitable for installation in hazardous locations unless housed in an enclosure suitable for the specified Division (or Zone), Class and Group(s).

Multiple Motors on a Single VFD

Special considerations are required when multiple motors are powered from a single VFD unit. Most VFD manufacturers can provide guidelines for proper motor thermal considerations and starting/stopping of motors. Cable runs from the VFD and each motor can create conditions that will cause extra stress on the motor winding. Filters may be required at the motor to provide maximum motor life.

Grounding and Cable Installation Guidelines

Proper output winding and grounding practices can be instrumental in minimizing motor related failures caused by PWM waveform characteristics and installation factors. VFD manufacturers typically provide detailed guidelines on the proper grounding of the motor to the VFD and output cable routing. Cabling manufacturers provide recommended cable types for PWM installations and critical information concerning output wiring impedance and capacitance to ground.

Integrated Motor and Inverter

By integrating the motor and inverter at NMC's manufacturing facility, many of the motor compatibility problems are minimized or eliminated. During the manufacturing process, the motor is matched to the inverter characteristics which ensures the winding temperature and torque levels meet the design specification. Since the inverter output wiring to the motor is nearly eliminated, bearing currents are rarely experienced. When the unit is properly grounded, reducing the output cable lengths in conjunction with an inverter grade insulation system and low factory setting of the switching frequency of the inverter drive, results in low risk of voltage peaks produced by the PWM waveform.

Vertical Motors on VFDs

Vertical motors operated on VFD power present unique conditions that may require consideration by the user or installation engineer:

- Locked rotor and drive tripping caused by non-reversing-ratchet operation at low motor speeds. It is not recommended to operate motors at less than 1/4 of synchronous speed. If slow speeds are required contact NMC engineering.
- Unexpected / unacceptable system vibration and or noise levels caused by the torque pulsation characteristics of the PWM waveform, a system critical frequency falling inside the variable speed range of the process or the added harmonic content of the PWM waveform exciting a system component
- Application related problems related to the controlled acceleration/deceleration and torque of the motor on VFD power and the building of system pressure/ load.
- The impact the reduction of pump speed has on the down thrust reflected to the pump motor and any minimum thrust requirements of the motor bearings
- Water hammer during shutdown damaging the non-reversing ratchet

Humidity and Non-operational Conditions

The possible build-up of condensation inside the motor due to storage in an uncontrolled environment or non-operational periods in an installation, can lead to an increased rate of premature winding or bearing failures when combined with the stresses associated with PWM waveform characteristics. Moisture and condensation in and on the motor winding over time can provide tracking paths to ground, lower the resistance of the motor winding to ground, and lower the Corona Inception Voltage (CIV) level of the winding.

Proper storage and maintenance guidelines are important to minimize the potential of premature failures. Space heaters or trickle voltage heating methods are the common methods for drying out a winding that has low resistance readings. **Damage caused by these factors are not covered by the limited warranty provided for the motor unless appropriate heating methods are properly utilized during non-operational periods and prior to motor start-up.**

NEMA[®] Application Guide for AC Adjustable Speed Drive Systems: <http://www.nema.org/stds/acadjustable.cfm#download>

* This information applies only to Integral Horsepower (IHP) motors as defined on the Agency Approval page, under UL[®] & CSA[®] listings where indicated.

Warranty Guidelines for Integral Horsepower (IHP)* Motors on Variable Frequency Drives

Warranty Guidelines

The information in the following section refers to the motor and drive application guidelines and limitations for warranty.

Hazardous Location Motors

Use of a variable frequency drive with the motors in this catalog, intended for use in hazardous locations, is only approved for Division 1, Class I, Group D hazardous location motors with a T2B temperature code, with a limitation of 2:1 constant torque or 10:1 variable torque output. **No other stock hazardous location motors are inherently suitable for operation with a variable frequency drive.** If other requirements are needed, including non-listed Division 2, please contact your Nidec Motor Corporation territory manager to conduct an engineering inquiry.

575 Volt Motors

575 volt motors can be applied on Inverters when output filters are used. Contact the drive manufacturer for filter selection and installation requirements.

Applying INVERTER GRADE® Insulated Motors on Variable Frequency Drives (2, 4, 6 pole)

The products within this catalog labeled "Inverter Duty" or "Vector Duty" are considered INVERTER GRADE® insulated motors. INVERTER GRADE® motors exceed the NEMA® MG-1 Part 31 standard. Nidec Motor Corporation provides a three-year limited warranty on all NEMA® frame INVERTER GRADE® insulated motors and allows long cable runs between the motor and the VFD (limited to 400 feet without output filters). Cable distance can be further limited by hot and humid environments and VFD manufacturers cable limits. These motors may be appropriate for certain severe inverter applications or when the factors relating to the end use application are undefined (such as spares).

Nidec Motor Corporation's U.S. Motors® brand is available in the following INVERTER GRADE® insulated motors:

- Inverter Duty NEMA® frame motors good for 20:1 Variable Torque & 5:1 Constant Torque, including Vertical Type RUSI (10:1 V.T.)
- Inverter Duty motors rated for 20:1 Constant Torque
- ACCU-Torq® and Vector Duty Motors with full torque to 0 Speed or 5000:1
- 841 Plus® NEMA® Frame Motors

Applying Premium Efficient motors (that do not have INVERTER GRADE® insulation) on Variable Frequency Drives (2, 4, 6 pole)

Premium efficient motors without INVERTER GRADE insulation meet minimum NEMA® MG-1, Section IV, Part 31.4.4.2. These motors can be used with Variable Frequency Drives (with a reduced warranty period) under the following parameters:

- On NEMA® frame 447 and smaller motors, 20:1 speed rating on variable torque loads & 4:1 speed range on constant torque loads.
- On TITAN® 449 and larger frame motors, 10:1 speed rating on variable torque loads.

- On TITAN® frame motors, inquiry required for suitability on constant torque loads.

Cable distances are for reference only and can be further limited by hot and humid environments (refer to Table 1). Refer to specific VFD

Table 1 - Cable Distances			
Maximum Cable Distance VFD to Motor			
Switching Frequency	460 Volt	230 Volt	380 Volt
3 Khz	127 ft	400 ft	218 ft
6 Khz	90 ft	307 ft	154 ft
9 Khz	73 ft	251 ft	126 ft
12 Khz	64 ft	217 ft	109 ft
15 Khz	57 ft	194 ft	98 ft
20 Khz	49 ft	168 ft	85 ft

manufacturers cable limits. Refer to the Motor/ Inverter Compatibility page for special consideration of vertical motor bearings.

Warranty Period Clarifications and Exceptions

Standard Energy Efficient Exclusion

Applying Standard & Energy Efficient Motors on Variable Frequency Drives is not recommended. VFD related failures on standard and energy efficient motors will not be covered under warranty.

Vertical Motor Windings

Premium efficient vertical motors without INVERTER GRADE® insulation that are installed using the criteria described in this document and applied in the correct applications shall have a warranty while powered by a VFD for 12 months from date of installation or 18 months from date of manufacturing whichever comes first. See limited warranty page for horizontal motor warranty periods.

Bearing Exclusion for Thrust Handling Bearings

Bearings used in premium efficient vertical motors, and all thrust handling bearings, that are powered by VFDs without shaft grounding devices or insulated bearings (when required) will not be covered under any warranty for damages caused from being powered by a VFD. All other bearing failure is covered per NMC's standard warranty. An electric motor repair shop approved to service U.S. MOTORS® brand motors must verify that the cause of the bearing failure was not due to Electrical Discharge Machining.

Medium Voltage and Slow Speed Considerations

Motors that are rated above 700 VAC or that are eight pole and slower require special consideration and installation and are not covered under the warranty guidelines in this document. Motors that are rated above 700VAC have special cable length and voltage differential issues that are specific to the VFD type and manufacture. The motor construction and cost may vary dramatically depending on the VFD topology and construction. Contact your NMC representative with VFD manufacturer name and model type for application and motor construction considerations. Motors that are designed eight pole and slower also require special installation and filters per the drive manufacturer.

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